

DISSERTATION

GEOGRAPHIC VARIATION IN LODGEPOLE PINE IN THE CENTRAL ROCKY MOUNTAINS

Submitted by

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WE HEREBY RECOMMEND THAT THE DISSERTATION PREPARED UNDER OUR
SUPERVISION BY Musser B. Moore ENTITLED Geographic Variation in
Lodgepole Pine in the Central Rocky Mountains BE ACCEPTED AS FULFILLING
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Abstract of Dissertation

GEOGRAPHIC VARIATION IN LODGEPOLE PINE IN THE CENTRAL ROCKY MOUNTAINS

Lodgepole pine (Pinus contorta Doug. ex Loud.) is one of the widest ranging conifers in North America, and is one of the most variable. The species has been taxonomically classified several times into from one to five distinct taxa. Currently, a scheme consisting of three varieties is accepted, in general.

This study reported on an investigation of phenotypic, geographic variation in Pinus contorta var. latifolia Engelm. One-hundred fifty-three individual sample trees from 34 natural populations which represented the latitudinal and elevational range of the species in Montana, Wyoming, and Colorado were included in the study.

Results were interpreted to show statistically continuous variation between populations for most of the 15 traits measured, but great within-population variation was also found.

Results from this study were compared to similar studies on the closely related species, jack pine (Pinus banksiana Lamb). Both lodgepole and jack pine may be young species with a common ancestor, and do not appear to have diverged far from each other in an evolutionary context. The possible evolutionary relationship of jack pine to lodgepole pine was discussed.

This study formed part of a larger project which was designed to develop a better understanding of the genetic and geographic variation of lodgepole pine in the central Rocky Mountains.

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Chapter I. Introduction

The nature of genetic recombination ensures that individual members of a given species are not genetically identical to each other, and often show phenotypic distinctiveness, as well. Within a species, the differentiation of component natural populations may or may not be sharp.(Mayr, 1977).

Unlike the individual, a population is dynamic with respect to genetic make-up and can respond to selection pressures over time through evolution. It is the evolution of populations which results in natural variation within a species. When the differences among populations within a species are considered in a spatial context, the term "geographic variation" is used to describe the phenomenon (Mayr, 1977).

Natural variation in plant populations relative to differences in geography was recognized and studied in the late nineteenth century (Darwin, 1859), and variation within a plant species due to environmental diversity in its range was systematically studied soon after (Turesson, 1922). Since then, methods and facilities used to evaluate geographic variation have greatly improved.

The importance of understanding geographic variation, particularly with regard to traits of economic value, has recently been stressed (Callahan, 1961; Haddock, 1967; Yeatman, 1976). Many morphological characteristics are highly correlated with adaptive,

physiological traits. The study of phenotypic variation between and within natural populations through systematic laboratory measurements and field study can give valuable insight for understanding geographic variation (Callahan, 1961; Barnes, 1967). This study was concerned with the geographic variation in natural populations of lodgepole pine (*Pinus contorta* var. *latifolia* Engelm.) which were sampled from Montana, Wyoming, and Colorado.

Natural Distribution and Ecology

Lodgepole pine has the most extensive latitudinal and elevational range of any tree species in North America (Critchfield, 1957; Critchfield and Little, 1966; Mirov, 1967; Pfister and Daubenmire, 1975). The species ranges from sea level along the Pacific coast to 3350 meters in Colorado, and from 63° north latitude near Stewart River in the Yukon to 31° north latitude in the southern Sierras. In Colorado, the southern limit of the species appears to be near La Veta Pass, 37° north latitude (Figure 1).

Lodgepole pine is generally considered to be a pioneer species, with the ability to rapidly regenerate disturbed sites, particularly after fire. Even 7000 years ago, coastal lodgepole pine seems to have been an initial tree invader on stabilized dunes (Mirov, 1967) and on infertile glacial depositions in the southern Oregon Cascades (Franklin and Dyrness, 1973).

Overall, the species shows an extremely broad ecologic amplitude with greater tolerances to environmental extremes than any other western conifer (Pfister and Daubenmire, 1975). The lodgepole pine

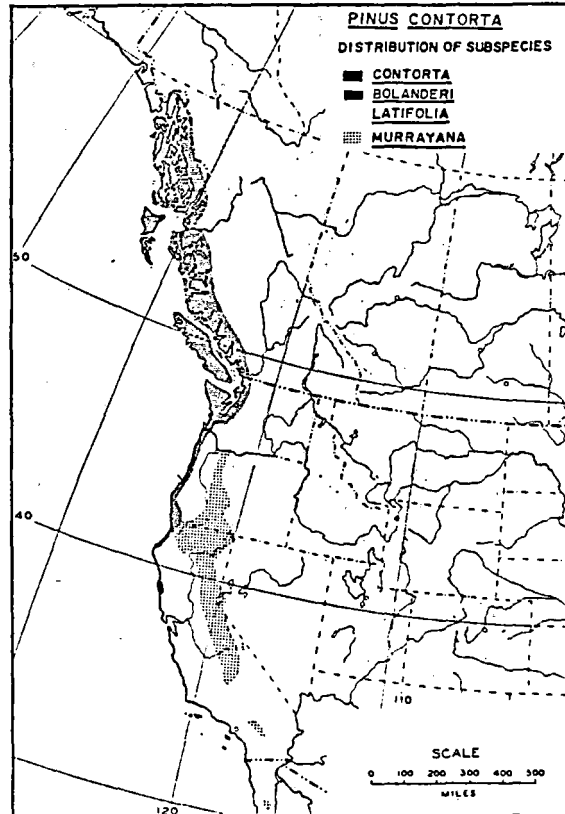


Figure 1. Approximate geographic distribution of lodgepole pine (from Critchfield, 1957).

forest cover type may occur on granite, shale, or sandstone derived soils in addition to calcareous till and pumice (Pfister and McDonald, 1980). In the Sierras and Cascades, poorly drained areas may support lodgepole pine where ponderosa pine (Pinus ponderosa Dougl. ex Laws.) would otherwise be successful (Fowells, 1965). In the Mendocino White Plains of California, lodgepole pine occurs on soils with a pH as low as 2.8 to 3.9 (Critchfield, 1980).

Four successional roles for lodgepole pine have been outlined by Pfister and Daubenmire (1975): minor seral, dominant seral, persistent, and climax. As a minor seral species, lodgepole pine is a component of mixed conifer stands and is generally replaced by more tolerant species within 100 years. The dominant seral role is most characteristic of lodgepole pine where it maintains dominance with Engelmann spruce (Picea engelmannii Parry ex Engelm.) and subalpine fir (Abies lasiocarpa (Hook.) Nutt.) at higher elevations and Douglas-fir (Pseudotsuga menziesii (Mirb.) Franco) at lower elevations for 100 to 200 years before being replaced. The rapid early growth of lodgepole pine leads to early dominance, but its shade allows initiation of the climax species. In the persistent role, the maintenance of lodgepole pine may be due to a lack of seed source for shade tolerant species, too great a stand density for invasion, or light surface fires. Persistent stands of lodgepole pine may be found on deep soils and gentle slopes in the upper elevations of the Colorado Front Range (Moir, 1969). Climax lodgepole pine stands are a result of edaphic or topoedaphic conditions. Two climax situations for lodgepole pine have been reported in the pumice region of Oregon: edaphic climax stands on poorly drained soils or topoedaphic climax

stands in frost pockets where cold spring temperatures do not permit survival of young ponderosa pine seedlings (Franklin and Dyrness, 1973).

Fowells (1965) classified lodgepole pine stands into either sub-climax (similar to "persistent", above), brought about initially by fire disturbance, or transitional, where an appreciable amount of advance reproduction of other species and a mixed overstory composition was present.

Evolutionary Origins

The genus Pinus was evidently distinct during the Cretaceous period and widely distributed throughout North America by the start of the Tertiary period, some 75 million years ago (Mirov, 1967). A fossil cone of Pinus pre-murrayana Knowlton, found in sediment near West Yellowstone, Montana, was apparently of Eocene epoch origin, but the dating of the specimen has been questioned (Critchfield, 1980; Mirov, 1967). During the Eocene (about 60 million years ago), the location of the Rocky Mountains was marked only by a low, north-south ridge. Topoedaphic modification of the environment in western North America was minimal at the time (Daubenmire, 1978). Yeatman suggested that both lodgepole pine and jack pine (Pinus banksiana Lamb.) originated from a common ancestor in the cool-temperate climate of higher latitudes at about the time of the Eocene, and that as the glacial epochs of the Quaternary approached and the uplift of the Rockies progressed, the two species differentiated. A cone from the

late Tertiary Faraday flora near Portland, Oregon was the first positive identification of ancestral lodgepole pine (Critchfield, 1980).

During the glaciation of the Pleistocene, the entire range of jack pine was covered with ice. Jack pine probably escaped extinction in refugia centered on the Appalachian highlands of eastern North America, reoccupying its former range through post-glacial migration to the northeast and northwest (Yeatman, 1967; Schoenike, 1976).

Lodgepole pine was evidently not widespread in western North America until after the Pleistocene glaciers receded. Reoccupation of the northern part of its range most likely occurred from populations to the south of the receding ice sheet (Critchfield, 1978). Early dominance of lodgepole pine in post-glacial succession in the Rocky Mountains is evidenced by large pollen deposits (Critchfield, 1978; 1980).

The occurrence of lodgepole pine in the Black Hills of South Dakota could indicate a more extensive range of the species which may have been broken up by the development of the Great Plains grasslands (Mirov, 1967). However, lodgepole pine in the Black Hills shows clear separation into three age classes, conspicuously corresponding to major fire events and Indian migrations (Rogers, 1969). There is no indication of glaciation in the Black Hills, and evidence for the glacial relict origin of the species in the area is scant.

Taxonomy and Past Studies on Variation

Lodgepole pine was collected by David Douglas near Cape Lookout, Oregon in 1825 and again near Cape Disappointment, Washington in 1826 (Mirov, 1954). Mertens also collected samples of lodgepole pine on Sitka Island, Alaska in 1826 (Mirov, 1954). Mertens' material was initially described as Pinus inops (Aiton) by Bongard in 1833, but both Douglas' and Mertens' material was ascribed to a new species described by Loudon as Pinus contorta Loudon in 1838 (Mirov, 1954; Critchfield, 1957). Since its first description, lodgepole pine has undergone several taxonomic re-classifications. Over the past century, the species has been divided up into as many as five taxonomic varieties or subspecies (Critchfield, 1978). Lodgepole pine has also been combined with jack pine into a single species, as well (Boivin, 1966). The confused state of the taxonomy of lodgepole pine has not entirely been resolved at the present.

In the late 1800's, Engelmann considered lodgepole pine to be dimorphic with an inland variety (Pinus contorta var. latifolia) and a coastal variety (Pinus contorta var. contorta) (Critchfield, 1957). The two-variety status of lodgepole pine was a popular view through much of the early and mid twentieth century, but Little (1953) considered the species to have no sub-specific categories. A revised version of Little's checklist (Little, 1979) describes three commonly accepted varieties of lodgepole pine at present: Pinus contorta var. contorta Dougl. ex Loud. (coast pine), Pinus contorta var. latifolia Engelm. (interior lodgepole pine), and Pinus contorta var. murrayana (Grev. and Balf.) Engelm. (Sierra lodgepole pine).

Roche (1966) described two variants of lodgepole pine on the basis of needle morphology, stem form, and bark features. The coastal form had heavy, numerous branches and dense, bushy crowns. The inland type had narrow crowns with sparse, light branches. Roche concluded from seedling studies that the form of lodgepole pine was largely environmentally modified. Jeffers and Black (1963) also described lodgepole pine as composed of two basically different types of populations on the basis of multivariate analysis of needle, female bud, and cone characteristics. Jeffers and Black included provenances from British Columbia, northwest Washington, Idaho and southern Oregon in their study. Critchfield has argued for at least four sub-specific categories for lodgepole pine, with a fifth race possibly warranting taxonomic status (Critchfield, 1980; Fosberg, 1959; Yeatman and Teich, 1969). Critchfield's work was based on variability in leaf and cone characteristics of natural populations and provenance tests including seed source material from throughout the species range. Critchfield's categories are summarized below (Critchfield, 1978):

Pinus contorta ssp. contorta Douglas ex Loudon

(Pacific coastal race)

Small, thick-barked, branchy trees; needles shorter and narrower than inland races; cones strongly reflexed.

Pinus contorta ssp. bolanderi (Parl.)

(Mendocino White Plains race)

A small population on highly acid soil of coastal Mendocino County, California. The trees have short, narrow needles lacking resin canals. Cones are reflexed and heavy.

Pinus contorta ssp. murrayana (Balf.)

(Sierra-Cascade race)

Trees of this race have short but broad needles. Cones are not serotinous and not persistent. The cones are also light in weight and have generally flat apophyses.

Pinus contorta ssp. latifolia (Engelmann ex Watson)

(Rocky Mountain - Intermountain race)

This is the most extensively distributed race of lodgepole pine. The cones are often serotinous and can be reflexed, upright, or semi-erect on the branch. Cones are normally hard and knobby. The needles of this race are the longest of the species, and three-needed fascicles occur in some of the more northern populations.

Del Norte race

This race is limited to serpentine soils in the low, coastal mountains of Del Norte County, California. The cones are the heaviest and most reflexed of any in the species and are often serotinous.

Interest in the geographic variation of lodgepole pine in the context of tree improvement work has not been as great as with more economically valued western conifers such as ponderosa pine (Read, 1980; Squillace and Silen, 1962; Rehfeldt, 1980), Douglas-fir (Rehfeldt, 1978), or western white pine (Pinus monticola Dougl. ex D. Don) (Steinhoff, 1979; Barnes, 1967). Although the basic biology and silvics of lodgepole pine did receive early study (Mason, 1915), until the 1950's seed source variation in lodgepole pine was considered important mostly to Europeans who had successfully introduced the species as a valuable exotic (Feilberg, 1964; Jeffers and Black, 1963). Lodgepole pine has also been introduced successfully to New Zealand (Cunningham and Roberts, 1970).

As lodgepole pine has gained recognition in North America as a valuable tree in terms of scenic, reclamation, timber, and watershed enhancement values, studies on geographic variation of the species have become more numerous. In British Columbia, a tree improvement program for lodgepole pine, based on the study of geographic and genetic variation in the species, was designed to: 1) delineate broad patterns of geographic variation, 2) identify populations outstanding in some desired trait, 3) formulate seed transfer rules, and 4) provide for the conservation of an array of population samples (Illingworth, 1970; 1971; 1973; 1975a; 1975b). The British Columbia studies have shown a substantial amount of variation within the species, and clinal variation for many traits along latitudinal and elevational gradients. Provenance studies which were started in 1967 have shown clinal variation in stomatal density (decreases south to north for coastal populations), autumn coloration (deepest color in northern populations), height growth (low elevation and low latitude sources grow for a longer period than high elevation and high latitude sources), and needle to seedling height ratio (in mid-summer, northern and high elevation sources showed short shoots and long needles; the ratio of needle length to seedling height increased south to north and from the coast inland) (Illingworth, 1970; 1971; 1975). Clinal variation patterns of variation in morphological traits of the closely related species, jack pine, have also been documented (Canavera and Wright, 1973; Hyun, 1977; Rudolf, Libby and Pauley, 1957; Schoenike, 1976; Sprackling and Read, 1975).

Few, if any, studies have adequately sampled and evaluated the natural variation in populations of lodgepole pine in the southern part of its range in the Rocky Mountains. A knowledge of the patterns of variation is essential to any tree improvement program for this species in this region. Basic information on the geographic variation of lodgepole pine can immediately be used to define guidelines for weed and seedling transfer, and to give insight to the interpretation and understanding of evolutionary trends and genetic structure of populations (Callahan, 1961; Yeatman, 1976).

This study consisted of two main parts: 1) An investigation of the phenotypic, geographic variation of traits of sampled parent trees of lodgepole pine and 2) a series of provenance tests established from seedling material of the parent trees. The objective of the parent tree morphology study, the results of which are presented in this report, was to sample and evaluate the phenotypic variation in traits of central Rocky Mountain populations of lodgepole pine, and to use the information in support or refutation of hypothesized continuous patterns of variation among natural populations. The objective of the provenance tests was to compare the variation in growth, survival, and phenology of lodgepole pine progeny from parent trees which represent the latitudinal and elevational range of the species in the central Rocky Mountains. The provenance tests are part of a long-term study, only the first-year progeny performance under greenhouse conditions was available for analysis in this report.

Chapter II. Materials and Methods

Tree species which have extensive geographic ranges often show patterns of phenotypic variation which coincide with geographic and climatic gradients in a gradual manner (Langlet, 1959). Because two strong determinants of local climate in the Rocky Mountains are elevation and latitude, these determinants might also influence the differentiation of natural populations of lodgepole pine. To assure a representative sample from an ordered gradient, this study used a systematic sampling procedure as suggested by Mendenhall, et al (1971) and Snedecor and Cochran (1967), that was defined by latitude and elevation.

Beginning at La Veta Pass, Colorado (latitude 37.6° N), the southernmost accessible stand of lodgepole pine in this study, and continuing to Marias Pass, Montana (latitude 48.3° N) stands of lodgepole pine were sampled from late July to early September, 1979. The original plan called for establishment of elevational transects at 1° latitudinal intervals in which the lowest, highest, and intermediate stands of lodgepole pine would be sampled at each location. Accessibility and time constraints modified the actual location of the natural population samples (Figures 2 and 3).

For nearly all of the sampled populations, five trees were selected for measurement and for collection of needle and cone material. Previous studies indicated that five trees per stand was

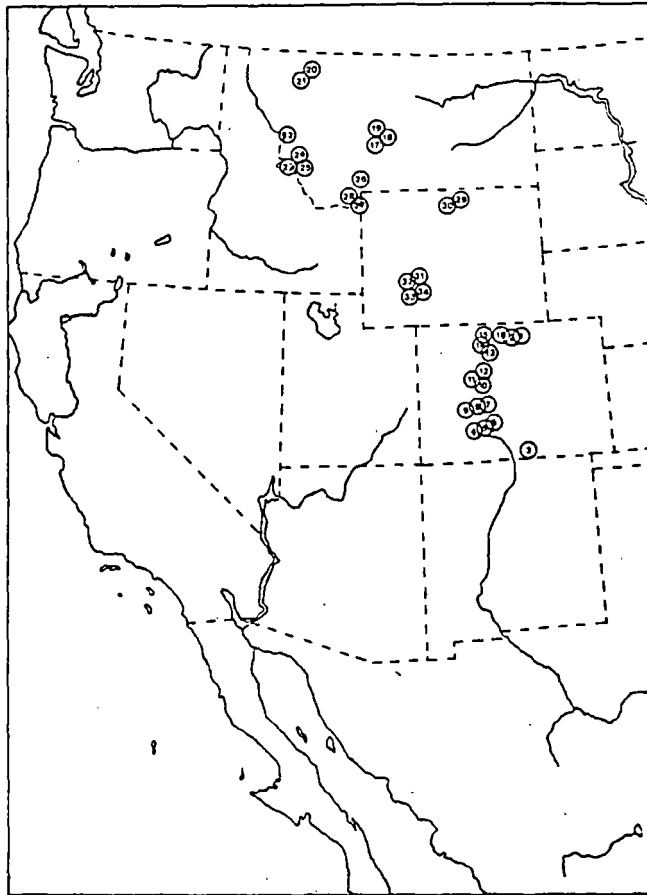


Figure 2. Location of the 34 natural population samples. Each numbered population was represented by five individual tree seed sources, with exceptions noted in Table 45.

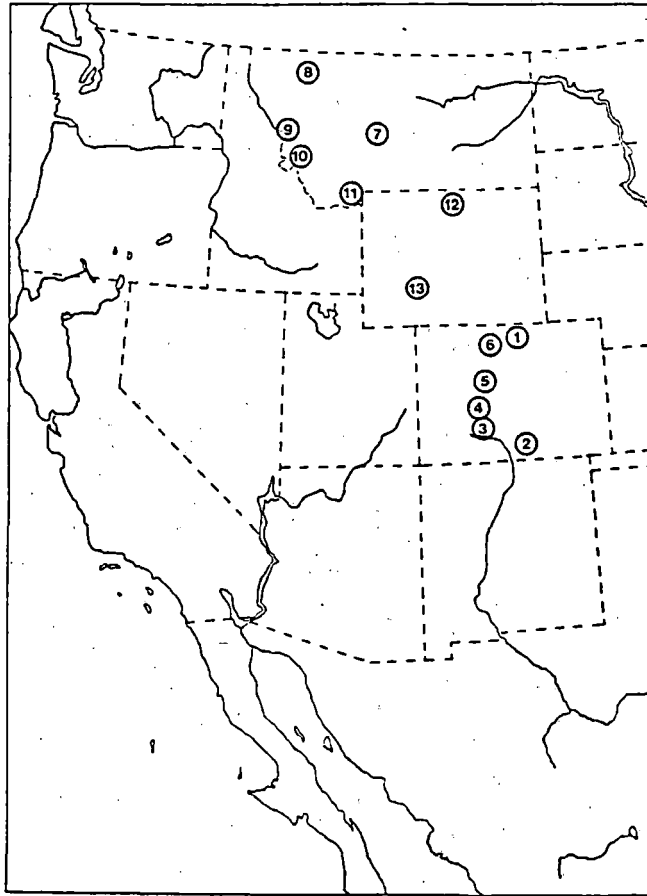


Figure 3. Identification of the 13 latitudinal locations. Each numbered location represents a composite of the natural populations (Figure 2) in that area. The number of natural populations and trees within natural populations contained within each latitudinal location is summarized in Table 45.

was an adequate sample to evaluate between and within population variability (Critchfield, 1957; Schoenike, 1976). If fewer trees were included in the population samples, the variance component between populations would be underestimated (Snedecor and Cochran, 1967). Including more trees would have limited the number of stands that could have been visited with the time constraints that were present.

Sampling of the individual trees in a stand was not random. The criteria of selection included: 1) tree dominance or co-dominance, 2) No obvious or severe insect or disease damage, and 3) the presence of seed cones. Individual tree sampling emphasized the healthy, breeding-age population which would be expected to contribute successfully to succeeding generations. The selected trees were generally 70 to 100 meters apart in a given stand. The distance between populations varied from 1 to 10 miles within a given latitudinal location. In all, 153 individual trees were sampled from 34 natural populations at different elevations within 13 latitudinal locations from La Veta Pass, Colorado to Marias Pass, Montana

Field Procedure

Site Characteristics

On field data sheets, latitude (degrees north) and longitude (degrees west) were recorded to the nearest 1/10 degree as determined from National Forest System and United States Geological Survey topographic maps. Elevation was recorded as meters above sea level. The topography of each collection site was classified as bottomland, slope, upland, or ridgetop and the percent slope of the terrain was

measured with a clinometer. Aspect was numerically classified from 1 to 8, with 1 corresponding to north, 2 to northeast, 3 to east, and so on. Seven categories of topsoil texture, based on visual inspection and touch, included gravel (1), sand (3), loam (5), and clay (7).

Sample Tree Measurements

For each tree sampled, diameter at 1.5 meters above the ground was measured to the nearest 0.1 centimeter with a diameter tape. Total tree height was measured with a clinometer to the nearest meter. Tree age was determined from an increment core taken at 1.5 meters above the ground.

Branch angle, the included angle subtended by the main stem axis of the tree and the axis of the branch, was measured on five mid-crown branches for each sampled tree. A mean value typified the tree. Branch angle was measured to the nearest degree by visually projecting a protractor held at arms length along the bole of the tree.

Several cone-bearing branches were cut from throughout the crown of each tree. The nature of cone orientation, abaxial (pointing away from the main axis of the tree) or adaxial (pointing towards the main stem axis), was noted:

Cone orientation angle was defined as the angle described by the tip of the cone, point of attachment to the twig, and twig apex. Cones with abaxial orientation had acute angles of orientation and adaxially oriented cones had obtuse angles (Figure 4). Cone orientation angle was measured on five cones from different branches with a protractor. A mean value typified the tree.

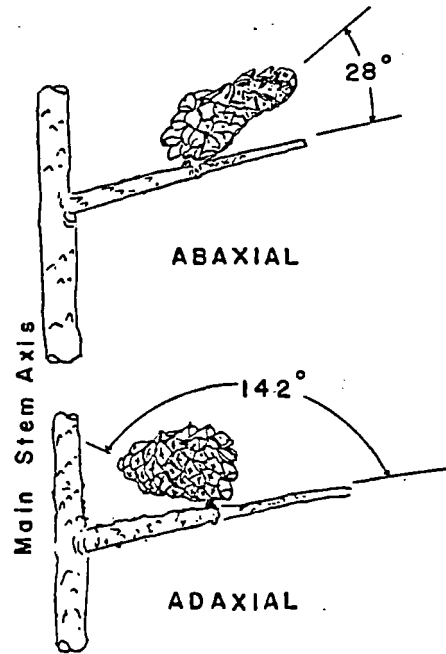


Figure 4. Method of measurement of cone orientation angle.

Needle persistence was measured on five different branches of each tree, with a mean value typifying a given tree. Beginning with the current year growth, internodes were counted back along the branch until an annual internode without needles was reached. The number of annual internodes with living needles equalled the needle persistence value in years for a given branch.

Estimates of cone serotiny were recorded according to five arbitrary categories ranging from 0 - 10 percent of the cones of a tree being serotinous (category 1) to 75 - 100 percent of the cones of the tree serotinous (category 5). Cone abundance for each tree was also estimated according to five categories from few cones (category 1) to many cones (category 5).

Cone and Foliage Samples

After sample tree measurements had been made in situ, cone and needle samples were collected. At least 25 seed cones from the previously cut branches were removed with pruning sheers and placed in paper sacks with identifying labels. From the same branches were taken samples of 10 needle fascicles from the mid-point of the previous season growth. The needle samples were transported back to the laboratory in small, plastic envelopes.

Stand Photographs

Before leaving each collection site, black and white photographs or Ektachrome slides were taken of the population. The photographs and associated notes are on file with the Department of Forest and Wood Sciences, Colorado State University. Table 1 summarizes the field measurements.

Table 1. Summary of field measurements

<u>Site Characteristics</u>		
<u>Feature</u>	<u>Units</u>	<u>Number of Observations</u>
latitude	degrees north	1 per site
longitude	degrees west	1 per site
elevation	meters a.s.l.	1 per site
topography	coded (see text)	1 per site
aspect	coded (see text)	1 per site
slope	percent	1 per site
topsoil	coded (see text)	1 per site
<u>Tree Measurements</u>		
<u>Feature</u>	<u>Units</u>	<u>Number of Observations</u>
tree height	meters	1 per tree
tree age	years	1 per tree
tree diameter	centimeters	1 per tree
branch angle	degrees	5 per tree
cone orientation angle	degrees	5 per tree
needle persistence	years	5 per tree
cone orientation	coded (see text)	1 per tree
cone serotiny	coded (see text)	1 per tree
cone abundance	coded (see text)	1 per tree

Laboratory Procedures

Seed Cones

Prior to physical measurement of the seed cones, seeds were extracted. Extraction of seed was carried out from October to December, 1979. When the cones were brought in from the field, they were transferred to screen-mesh containers. A hot-water bath (85°C) was prepared, and each screen container with cones was given a 30-second dip in the water to release the resinous bond of the serotinous cone scales. Once dipped in the water, the screen containers were hung at room temperature for the cones to dry, open, and release seed. In most cases, cones were fully open within three weeks of the hot water treatment. Cones from some seed sources needed a second water treatment to break the resin bond.

Released seeds were shaken from the screen containers into de-winging sieves, dewinged by hand, and run through a "Clipper" seed-cleaning machine before storing them in plastic envelopes at 0 - 4°C. Each of the 153 individual-tree seed lots was labeled and stored separately.

Morphological measurements of fully open cones were taken after seed extraction. Usually, several cones from a given seed source were measured and a mean value used to typify that source. Seed cone measurements included an evaluation of apophysis knobiness, cone length, cone width, cone shape, cone volume, cone weight, and cone specific gravity.

Five arbitrary apophysis shape categories, from flat (category 1) to attenuate (category 5) were used to classify each of five cones randomly selected from each of the 153 seed sources. A mean value typified a given source.

Cone length was measured to the nearest millimeter from the point of attachment to the twig to the tip of the cone. A mean value represented the best estimate for each seed source. Similarly, cone width was measured at the widest part of the cone when the scales were open. A derived value of cone shape, the ratio of length to width, was calculated for each cone measured.

Cone volume was measured on the same five cones measured for length, width, and shape. The volume to the nearest 0.1 ml was determined by the water-displacement method, described by Critchfield (Critchfield, 1957). A mean value was calculated for each seed source.

Oven-dry cone weight was determined to the nearest 0.1 gram by drying a sample of 20 cones for 48 hours at 105°C and obtaining a mean value for each seed source. Cone specific gravity was then derived by dividing the mean cone weight value by the mean cone volume value for each seed source.

Needle Measurement

The only needle measurement included in this study was average dry weight of a fascicle. For each seed source, 10 fascicles were oven-dried at 105°C for 48 hours prior to weighing. A mean value was calculated to typify a given seed source.

Laboratory measurements are summarized in Table 2.

Table 2. Summary of laboratory measured traits

<u>Cone Measurements</u>		
<u>Feature</u>	<u>Units</u>	<u>Number of Observations</u>
cone length	millimeters	5 per tree
cone width	millimeters	5 per tree
cone shape	ratio: length/width	5 per tree
cone volume	milliliters	5 per tree
cone weight	grams	1 per tree ¹
cone specific gravity	unitless	1 per tree ²
apophysis shape	coded (see text)	5 per tree
<u>Needle Measurement</u>		
<u>Feature</u>	<u>Units</u>	<u>Number of Observations</u>
needle biomass	milligrams	1 per tree ³

¹Mean cone weight based on a composite sample of 20 cones.

²Mean specific gravity derived as a ratio of mean cone weight and mean cone volume for a particular sample tree.

³Mean dry weight of a fascicle based on a composite sample of 10 fascicles.

Seedling Growth

In June 1980, seeds from the 153 individual-tree seed sources were sown at the Colorado State Forest Service nursery in the production greenhouse. Seeds were sown in plastic, containerized seedling tubes filled with a peat-perlite-vermiculite (1:1:1) potting soil mix. The seedlings were arranged according to a randomized block design, with each seed source represented with two plots of 42 seedlings each. The seedlings germinated and grew under the accelerated growth conditions of the greenhouse. These growth conditions resulted in unusually tall seedlings in some cases, and unpredictable periodicity in extension growth. A power failure on July 4, 1980 caused overheating and severe mortality throughout the planting; re-seeding could only be done for a few of the seed sources.

Given the uncontrollable irregularities, only the final, first season greenhouse height of the surviving seedlings was used to compare the seed sources. Measurements of 14 seedlings from each plot, if available, were taken in January 1981 after the accelerated growth conditions had been shut down, and the seedlings had hardened off. Seedling height was measured to the nearest millimeter, and a mean value for each plot was calculated and recorded.

After measurement, the seedlings were labeled and moved to a lath-house where they remained until outplanting. One outplanting which consists of a single randomized block with three replicates of each of the 153 seed sources was established at the Colorado State Forest near Gould, Colorado (40.5° N latitude, 106.0° W longitude, elevation 2674 meters) in June, 1981. Other plantations are planned.

Statistical Methods

Mean values for morphology and greenhouse measurements were analyzed using the SPSS (Statistical Package for the Social Sciences) package (Nie, 1975) available on the CDC-6400 computer at Colorado State University. Four basic analyses evaluated the variation between and within the natural populations. Analyses included Pearson product-moment correlations, stepwise multiple linear regression, one-way analyses of variance, and Duncan's multiple range test.

Correlation coefficients, calculated for every combination of field site characteristics, field tree measures, laboratory measures, and greenhouse measures, revealed the strength of the relationship between a given pair of variables. Stepwise multiple linear regressions developed a number of models relating the phenotypic measures to the geographic characteristics of the seed source locations. Analyses of variance tested the null hypothesis of no differences between the 34 natural populations or between the 13 latitudinal locations. To evaluate the patterns of population-to-population or location-to-location variation, Duncan's multiple range test (described in Nie, 1975) examined each analysis of variance which showed significant differences.

Chapter III. Results

Variation in sample tree traits are presented in tabular and graphical form in this section. Some traits were handled as continuous variables in the statistical analyses (branch angle, cone angle, cone length, cone width, cone shape, cone volume, cone weight, cone specific gravity, needle biomass and seedling height), whereas other traits were treated as discrete variables (class value traits such as needle persistence, cone orientation, cone serotiny, cone abundance, and apophysis shape). Multiple regression analysis was performed only on the continuous variable traits, but all traits were analyzed with correlation analysis, one-way analysis of variance, and Duncan's multiple range tests.

Table 3 gives the overall descriptive statistics for each sample trait. Tree height, diameter and age are included here to give an indication of the size and age of tree selected for this study. These three traits were considered to be under strong environmental control, and were useful in providing a general indication of the type of tree and stand sampled for this study. The large coefficients of variation associated with these three traits indicated that, considering all trees sampled for the study, a great variety of sizes and ages were included. The variation in size and age of sample trees could have introduced a bias into the sampling if, for instance, significantly taller trees were consistently sampled at higher elevations or, larger

Table 3. Overall descriptive statistics for sample tree data.

Trait	Mean ¹	Standard Deviation	Range	Coefficient of Variation
tree height (m)	10.9	2.5	4.0 - 19.0	23.0%
tree diameter (cm)	23.7	8.5	7.6 - 57.4	35.9
tree age (years)	83.2	39.8	20.0 - 240.0	47.8
branch angle (degrees)	72.4	12.5	30.0 - 95.0	17.2
needle persist. (years)	8.7	1.8	3.0 - indef.	20.8
cone orientation (class)	1.7	0.4	1.0 - 2.0	25.4
serotiny (class)	4.1	0.9	2.0 - 5.0	21.5
cone abundance (class)	3.0	1.0	1.0 - 5.0	32.9
cone length (mm)	37.7	4.3	29.4 - 50.2	11.3
cone width (mm)	35.9	4.5	25.4 - 50.8	12.5
cone shape (ratio)	1.1	0.1	0.9 - 1.4	8.9
cone volume (ml)	8.6	2.7	3.2 - 17.4	30.7
apophysis shape (class)	3.4	0.9	1.0 - 5.0	27.1
cone weight (gm)	5.6	1.6	1.8 - 9.9	28.3
cone specific gravity (percent)	67.1	10.2	42.0 - 89.0	15.2
needle biomass (mg)	3.5	1.4	2.0 - 6.1	39.3
cone angle (degrees)	106.3	24.1	40.0 - 135.0	22.6
greenhouse seedling ht. (mm)	53.3	16.3	23.0 - 91.6	30.6

¹For all traits, sample size equals 153 individual trees.

diameter trees were systematically sampled on flat terrains, and so on. Table 4 shows, however, very little relationship between tree height, diameter, and age and any of the seven recorded sample site characteristics. In summary, although the size and age of trees selected for the study varied considerably, that variation was apparently distributed over all of the various locations visited and sampled. A possible exception might be shown by the significantly high correlation between tree age and topography, which indicated that older trees were sampled at locations higher on hillsides.

For the remainder of this chapter, the results pertinent to each phenotypic trait will be presented separately and a short discussion of those results will accompany the summary tables and figures. General discussion of the results will follow in the next chapter.

Results of Traits Measured in the Field

Branch Angle

Average branch angle varied between 30° and 95° over all trees sampled and was significantly correlated with all seven sample site characteristics. In general, branch angle increased from south to north, increased east to west, decreased with an increase in elevation, increased with steeper slopes, and increased with more clay-like topsoil textures.

The geographic variability in branch angle is graphically represented in Figure 5. Similar figures were developed for all continuous variable traits, and the description of this figure generally

Table 4. Pearson product-moment correlation coefficients.

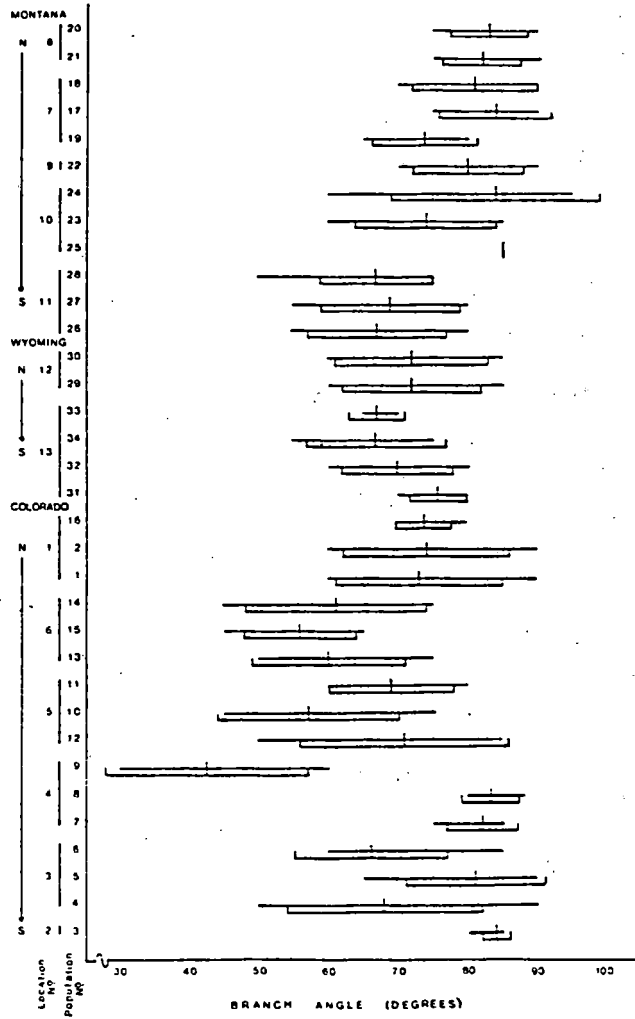
Phenotypic variable	Latitude	Longitude	Elevation	Topography	Aspect	Slope	Topsoil
tree height	.0255	-.0237	-.0366	.0004	-.0307	-.0332	.0910
tree diameter	.0932	-.0158	.0081	.1648*	-.0979	-.1958*	.0728
tree age	-.0398	-.1078	.1764*	.3262***	.1837**	-.0109	-.0788
branch angle	.2612***	.2869***	-.3141***	-.2064**	.1874*	.2912***	-.3922***
cone angle	.1229	.0635	.1887*	.0684	.1397	.1134	-.1258
cone orientation	-.1771*	-.0018	.2483**	.0669	.0945	.0776	-.0898
cone serotiny	.1198	.1198	-.1209	-.1711*	-.0108	-.1468	.0647
cone abundance	-.0217	-.2208**	.0475	.0567	-.0620	-.0551	.0672
needle persistence	-.4387***	-.4864***	.4718***	-.0804	-.0630	.1860*	.0822
cone length	-.2006*	-.3257***	.2451**	-.0211	-.0107	.0394	.0902
cone width	-.0210	-.0624	.0892	-.0117	-.0733	.0102	-.0802
cone shape	-.2461**	-.3380***	.2159**	-.0255	.0914	.0477	.1887*
cone volume	-.1673*	-.3388***	.2188**	.0657	-.0064	-.0485	.2054*
apophysis shape	-.0755	-.1326	.0410	-.0694	-.0948	-.0284	.0195
cone weight	.0805	-.1450	-.0115	-.0710	-.0932	.0944	.0554
cone specific gravity	.4755***	.3712***	-.4134***	-.0528	-.0320	.1440	-.2648***
needle biomass	.1843*	.1955*	-.0544	.2820***	-.0512	.1024	-.1018
greenhouse height	.1171	.1343	-.2285**	-.1516	.0413	.1192	-.1828

*Significant at the 5 percent level

**Significant at the 1 percent level

***Significant at the 0.1 percent level

Figure 5. Branch angle for sampled populations. Explanation of symbols is on pages 31 - 32.



holds for all continuous-variable traits. On the vertical axis of the figure are two columns of numbers: location numbers and population numbers (see Figures 2 and 3 in Chapter II for explanation of the features of population and latitudinal location designations). The location numbers are ordered from north to south, the northern-most location is at the top of the figure. Within each location, the component population numbers are ordered from highest elevation to lowest elevation with the high elevation populations at the upper end of a vertical line which connects all population numbers for a given location. Low elevation population numbers are at the lower end of each vertical line. A scale representing the range of values for the trait forms the horizontal axis of the figure. For each population, the mean, plus and minus one standard deviation, and range of measurements for the trait is shown. The mean is designated with a short vertical line, plus and minus one standard deviation is shown by a U-shaped bracket beneath and centered around the mean line, and the range of measurements is shown with a horizontal line running through the middle of the vertical mean bar. With this diagram, the relative variability between and within both populations and sample locations is visually apparent. Additionally, elevational and latitudinal trends (if any) can be seen. For instance, the significant correlation of branch angle with latitude is suggested by the trend to the upper right of the components of Figure 5.

Another feature of Figure 5, and others like it, is the possible illustration of discontinuities, or, geographic "groupings" of populations. One can visualize two or three geographically distinct segments of Figure 5. From south to north, populations 3 through 7 in

southern Colorado seem to form a unit, populations 9 through 28 form another, and populations 25 north through 20 yet another.

Analysis of variance results showed significant differences ($P < 0.05$) between populations on the basis of branch angle. Statistically distinct groups of populations are shown in Table 5. These statistical groupings of all 34 populations did not show any clear geographic trend, but the overlapping nature of the groups indicated that variation between the populations was continuous. This result was typical for all traits where significant differences between populations were found. Since graphing the results of Duncan's multiple range tests on 34 populations, which represented a variety of elevations and latitudes, would produce a cluttered map that would not easily be interpreted, Duncan's procedure was also run on the 13 latitudinal locations, and the results of this test could be reasonably mapped (Figure 6, Table 6)¹.

In order to evaluate how much of the variation in a phenotypic trait could be explained by a regression model composed of independent variables which were geographic features of the collection sites, stepwise multiple linear regression was performed on those phenotypic traits which could be treated as continuous dependent variables. Table 7 summarizes the results of regression analysis on branch angle.

¹ Reducing the number of components in a multiple range test when those components represent sample localities in a geographic variation study often gives more easily interpretable results and a better illustration of broad, geographic variation patterns. This technique was suggested by Dr. R. A. Read at the Rocky Mountain Forest Genetics Committee Meeting held in Denver, Colorado in March 1981.

Figure 6. Geographic pattern of statistically different groups of sample locations based on branch angle. Locations indicated by the same symbol(s) are not significantly different at the 5 percent level by Duncan's multiple range test. Symbol coding for this figure and others similar to it is as follows:

▴ Largest value for variable measured
▢
■
□
●
○ Smallest value for variable measured



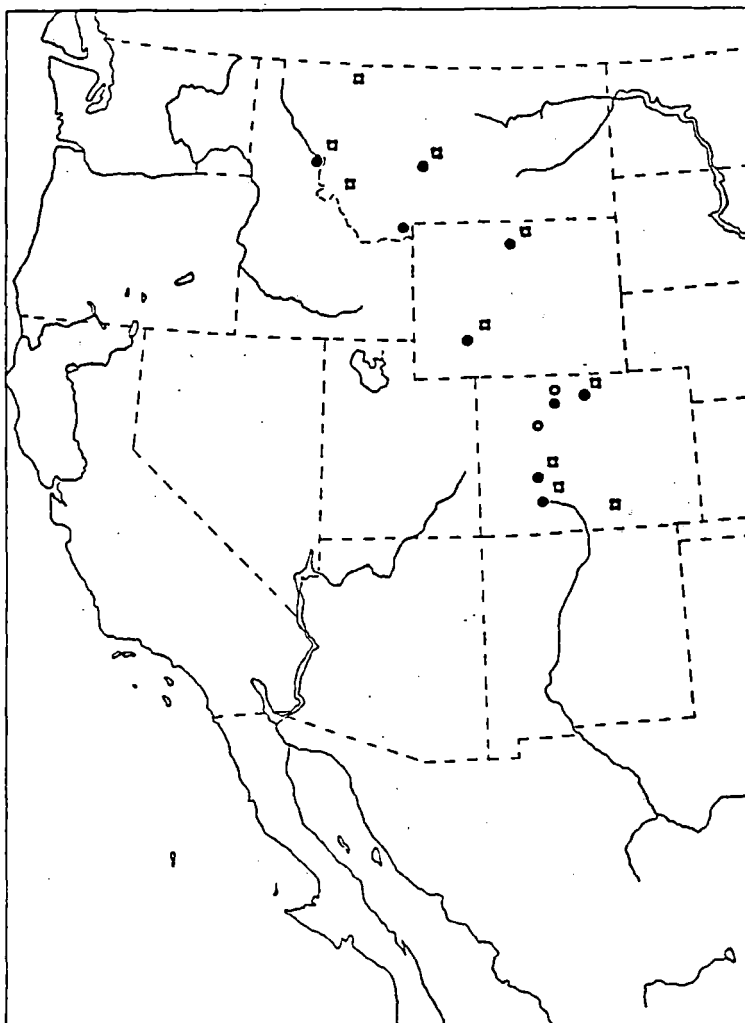


Table 5. Duncan's multiple range test on branch angle by population. Mean values spanned by a vertical line are not significantly different at the 5 per cent level.

Population Number	Mean Value (degree)
9	43.3
15	56.0
10	58.0
13	60.0
14	61.0
6	66.0
28	66.7
34	66.7
26	67.0
33	67.5
4	68.0
11	69.0
27	69.0
32	70.0
12	71.0
29	72.0
30	72.0
1	73.0
19	73.8
2	74.0
16	74.0
23	74.0
31	76.0
22	80.0
5	81.0
18	81.3
7	82.0
21	82.0
20	83.0
8	83.3
17	83.8
3	84.0
24	84.0
25	85.0

Table 6. Duncan's multiple range test on branch angle by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (degrees)
6	59.0
5	66.0
11	67.7
13	71.0
3	71.7
12	72.0
4	72.8
1	73.7
7	79.6
9	80.0
10	80.0
8	82.5
2	84.0

In Table 7, and other tables similar to it in this chapter, there is an "overall" regression model and a series of "location" models. The overall model is based on data from all 153 sample trees and the location models are based on data only from trees of a given latitudinal location. Latitude and longitude were not candidate independent variables in the location models since those values would have been generally the same for all trees of a given location.

Table 7 shows that, for branch angle, a model including topsoil texture, sample tree diameter, sample tree age, and elevation together accounted for 32 percent of the variability in the trait. With the exception of location 1, topsoil texture and sample tree diameter were important in explaining variation in the Colorado and Montana locations, and slope percent of the collection site was significant in the Wyoming location in explaining variation in branch angle.

Cone Orientation Angle

For all trees sampled, cone angle ranged from 40° to 153° and averaged 106° . Of the geographic variables included in this study, only elevation was correlated significantly to cone angle, overall: at higher elevations, cone angle was significantly greater than at lower elevations. Figure 7 shows much less variability in cone angle within populations at the southern end of the sampled distribution. Populations 3 through 12 in Colorado seem to form a group with low variability for this trait. Cone orientation angle is considered to be under strong genetic control in lodgepole pine

Table 7. Regression analysis on branch angle

Bas ¹	Equation ²	R ³	F ⁴	p ⁵
Overall	104.72 - 3.37(soil ⁶) - 0.05(dia) + 0.09(age) - 0.006(elev)	.32	(4,148): 17.190	< 0.001
Location 1 (CO-1, CO-2, CO-16)	No significant regression ⁷			
Location 3 (CO-4, CO-5, CO-6)	107.68 - 6.36(soil) - 0.81(dia)	.43	(2,12): 4.635	0.032
Location 5 (CO-10, CO-11, CO-12)	78.84 - 0.14(dia) + 2.92(ht) - 1.80(aspect)	.55	(3,11): 4.578	0.026
Location 7 (MT-1, MT-2, MT-3)	47.50 + 8.75(soil)	.26	(1,10): 3.589	0.087
Location 8 (MT-4, MT-5)	93.77 - 0.05(dia)	.32	(1,8): 3.780	0.088
Location 13 (WY-3, WY-4, WY-5, WY-5A)	55.64 + 1.07(slope)	.25	(1,13): 4.257	0.060

¹ "Overall" regression equation is based on observations from all 153 sample trees, regardless of latitudinal location. "Location" regression equations are based on observations from only those trees included in a given latitudinal location. Latitude and longitude were possible independent variables only in the "Overall" equation.

² The linear regression model describing the dependent variable.

³ The multiple correlation coefficient for the regression model.

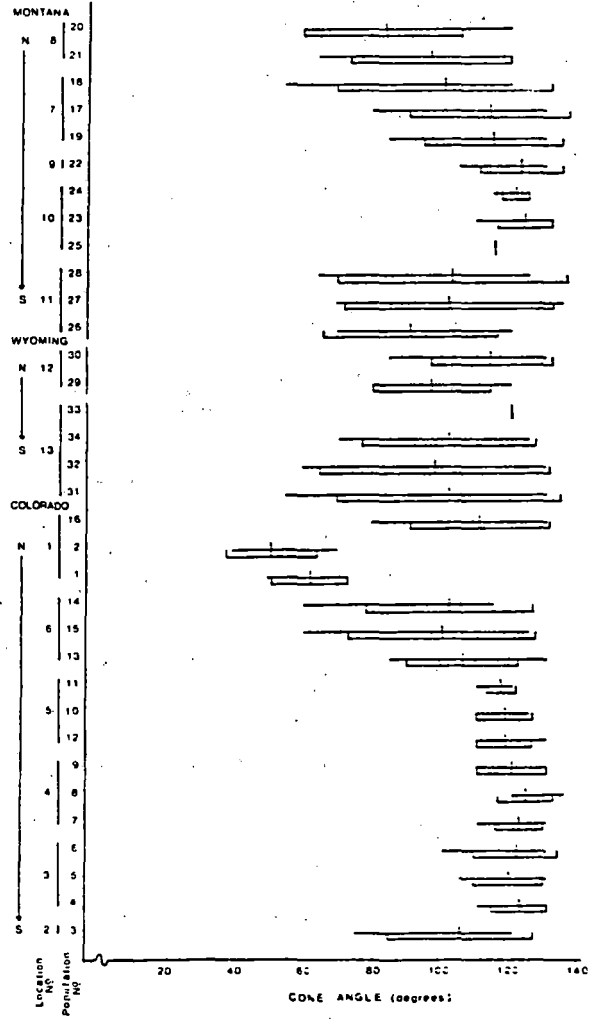
⁴ The F-ratio for the regression model. First number in parentheses is the regression degrees of freedom and the second number is the residual degrees of freedom. The number following the colon is the ratio of regression mean square to residual mean square.

⁵ Significance of the regression model in terms of the probability of obtaining the presented results as a result of random chance alone.

⁶ Independent variable code words are as follows: lat: latitude, long: longitude, elev: elevation, dia: sample tree diameter, ht: sample tree height, age: sample tree age, soil: topsoil texture, aspect: aspect of the sample site, slope: slope percent of the sample site, topo: topographic position on slope of sample.

⁷ The term "No significant regression" means that no independent variable had a regression coefficient significant enough to include in a regression model with the dependent variable.

Figure 7. Cone orientation angle for sampled populations. Explanation of symbols is on pages 31 - 32.



(Critchfield, 1980), and the possibility of jack pine influence on the expression of this trait in lodgepole pine through introgression has been suggested. This topic will be explored further in the next chapter, but the lower variation in this trait, and the higher values for cone orientation angle in the more southern collections, farther away from the natural range of jack pine, would not be in disagreement with a hypothesis for introgression of jack pine into lodgepole pine for this trait.

Significant between-population and between-latitudinal location variation was found for cone orientation angle, and the population-to-population and location-to-location variation appeared continuous (Table 8, Table 9). Broad, geographic patterns were not apparent for cone angle, except as noted above. A pocket of small cone orientation angle shows up in Figure 8 in the northern Colorado Front Range.

Table 10 indicates that 17 percent of the overall variability in cone orientation angle was explained by the model including elevation and longitude as independent variables. At individual locations, aspect and topsoil texture were site characteristics important in explaining the observed variation in cone orientation angle.

Cone Orientation

Cone orientation was a class-value trait, with only two possible classifications: abaxial (class 1) and adaxial (class 2). All cones of the same tree had the same classification, but there was often a mixture of both types of tree in a single population.

Table 8. Duncan's multiple range test on cone angle by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (degrees)
1	51.0
2	62.0
20	84.0
26	91.0
21	97.0
29	97.0
32	98.0
15	100.0
18	101.3
34	101.7
14	102.0
27	102.0
31	102.0
28	103.3
3	104.8
13	106.0
16	111.0
17	113.8
30	114.0
19	115.0
25	115.0
11	117.0
10	118.0
12	118.0
5	119.0
9	120.0
33	120.0
6	121.0
7	121.0
24	121.0
4	122.0
22	122.5
8	123.8
23	124.0

Table 9. Duncan's multiple range test on cone angle. by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (degrees)
1	74.7
8	90.5
11	98.1
6	102.7
13	103.0
2	104.8
12	105.5
7	110.0
5	117.7
3	120.7
10	121.3
4	121.7
9	122.5

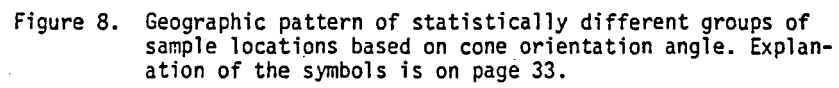


Figure 8. Geographic pattern of statistically different groups of sample locations based on cone orientation angle. Explanation of the symbols is on page 33.

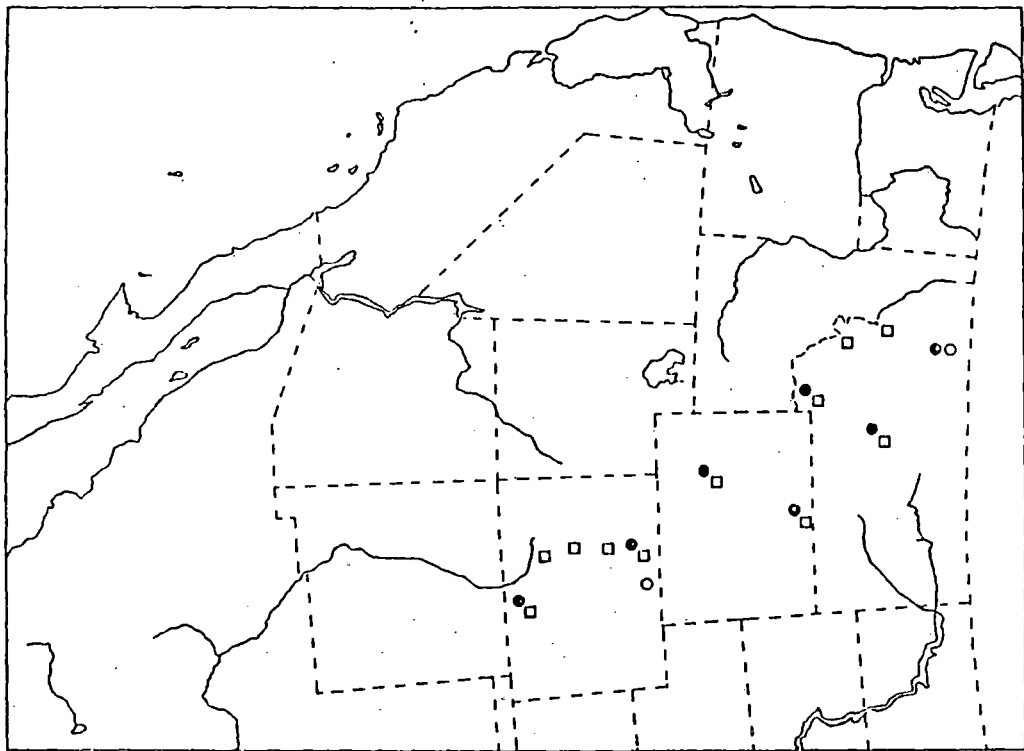


Table 10. Regression analysis on cone orientation angle.

Basis	Equation	R ²	F	P
Overall	- 487.26 + 0.03(elev) + 4.78(long)	.17	(2,150): 14.91	< .001
Location 1 (CO-1, CO-2, CO-16)	2.00 + 13.625(aspect)	.76	(1,13): 41.89	< .001
Location 3 (CO-4, CO-5, CO-6)	87.51 + 0.32(age) + 5.30(soil) - 0.08(dia)	.41	(3,11): 2.602	0.105
Location 5 (CO-10, CO-11, CO-12)	133.58 - 1.42(ht)	.23	(1,13): 3.605	0.080
Location 7 (MT-1, MT-2, MT-3)	No significant regression			
Location 8 (MT-4, MT-5)	205.07 - 1.20(age)	.17	(1,8): 1.669	0.230
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

A significant negative correlation of cone orientation with latitude indicated that the incidence of abaxial cone orientation was greater in northern collections as compared to southern collections. Additionally, a highly significant positive correlation of cone orientation with elevation suggested a greater incidence, overall, of adaxial cones at higher elevations.

Figure 9 shows a histogram of mean values for cone orientation by population. This type of figure was developed for all class-value traits. The vertical axis is identical to figures such as Figure 5. From Figure 9, it is apparent that the incidence of abaxial orientation generally increased from southern to northern populations.

As with most traits studied, significant differences were found between populations and locations with respect to cone orientation, and the multiple range tests indicated continuous variation patterns in Tables 11 and 12. Figure 10 shows, again, that adaxial orientation was typical for the southern-most locations sampled; the incidence of abaxial cones increased to the north.

Cone Serotiny

Cone serotiny was another class-type trait. Although there were five categories possible ranging from no serotinous cones to all cones serotinous, no tree was found to have all serotinous cones. Overall, 75 percent of the cones on most trees were closed in this study.

Of the recorded site characteristics, only topography was significantly correlated with serotiny. Higher levels of cone openness were associated with higher site elevations.

Figure 9. Cone orientation of sampled populations. Histograms correspond to population mean values.

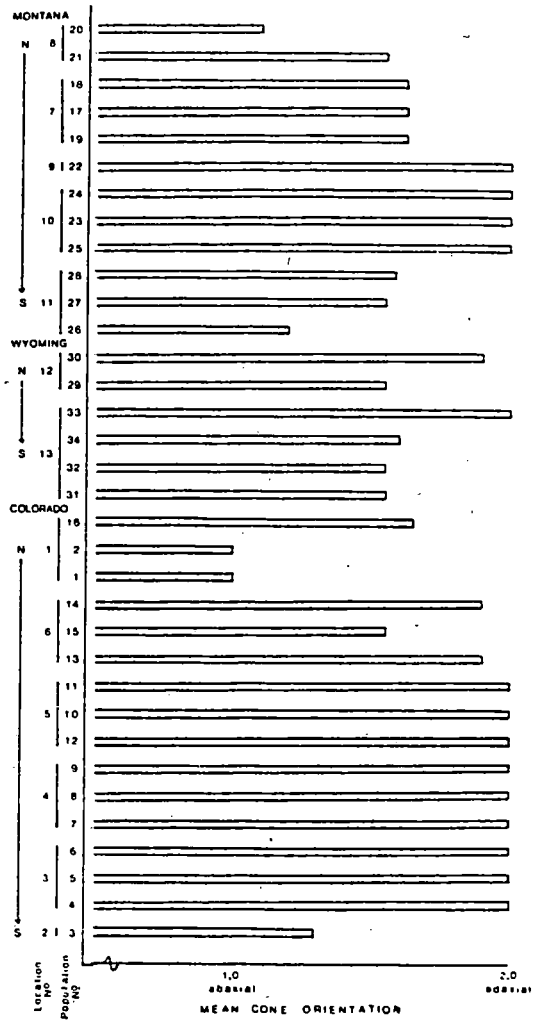


Table 11. Duncan's multiple range test on cone orientation by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (class value)
1	1.0
2	1.0
20	1.2
26	1.4
3	1.6
15	1.6
21	1.6
27	1.6
29	1.6
31	1.6
32	1.6
28	1.7
34	1.7
17	1.7
18	1.7
19	1.7
13	1.8
14	1.8
16	1.8
30	1.8
4	2.0
5	2.0
6	2.0
7	2.0
8	2.0
9	2.0
10	2.0
11	2.0
12	2.0
22	2.0
23	2.0
24	2.0
25	2.0
33	2.0

Table 12. Duncan's multiple range test on cone orientation by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (class value)
1	1.3
8	1.4
11	1.5
2	1.6
13	1.7
12	1.7
6	1.7
7	1.8
3	2.0
4	2.0
5	2.0
9	2.0
10	2.0

Figure 10. Geographic pattern of statistically different groups of sample locations based on cone orientation. Explanation of symbols is on page 33.

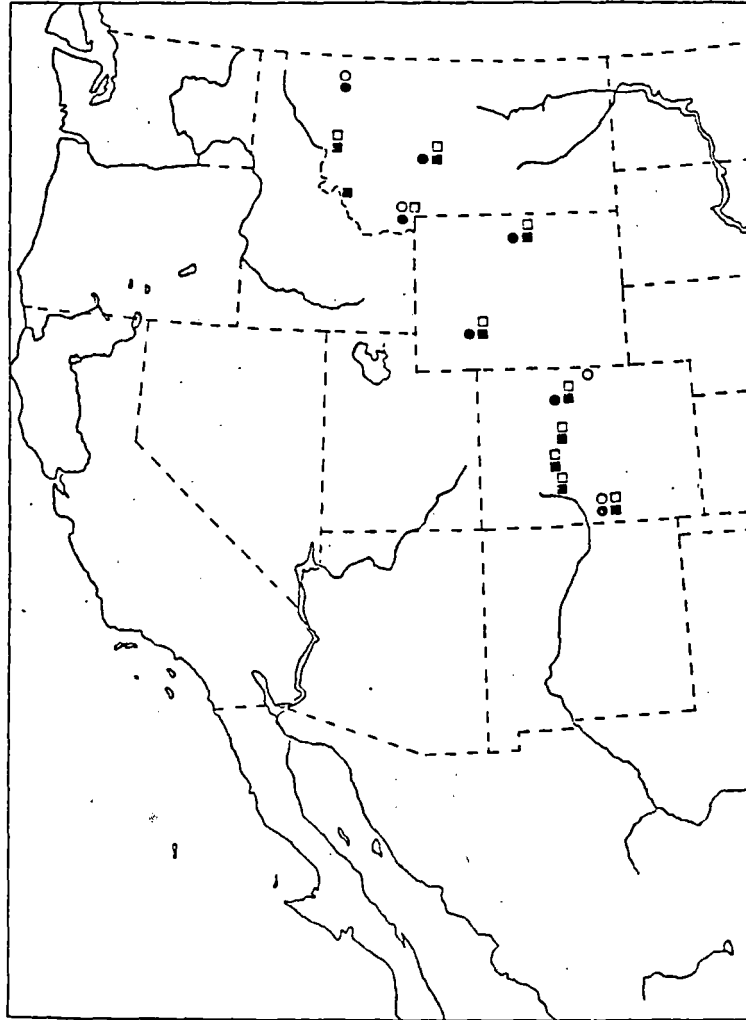


Figure 11 shows little in the way of geographic trends for population mean values of serotiny, but there might be an indication of a slight decrease in open-cone habit in the more northern populations. Differences between populations and locations were significant, but the population-to-population and location-to-location variation was continuous (Table 13, Table 14).

Table 14 and Figure 12 show a broad trend for greater incidence of open cones in the more southern samples. However, a pocket of open-cone habit did seem apparent in southwest Montana. Critchfield (1957) felt that serotiny varied clinally with elevation in lodgepole pine, and that more open cones were found at higher elevations. No such trend was clear in the current study, but further discussion of this feature follows in the next chapter.

Cone Abundance

Cone abundance was correlated most strongly with longitude, and varied negatively with that geographic feature. Cones were more abundant on sample trees in the eastern-most part of the sampled range than in the west. Figure 13 shows great variation among populations for this trait, but there appears to be two "troughs" of abundance, one approximately in the center of Colorado from population 8 north to population 14, the other in Montana near populations 22, 23, and 24. Peak cone abundance was in Wyoming populations 31 through 34.

Significant differences were found between populations and latitudinal locations, and the Duncan's multiple range tests showed the typical continuous, overlapping nature of the variation (Table 16, Table 17).

Figure 11. Cone serotiny of sampled populations. Histograms correspond to population mean values.

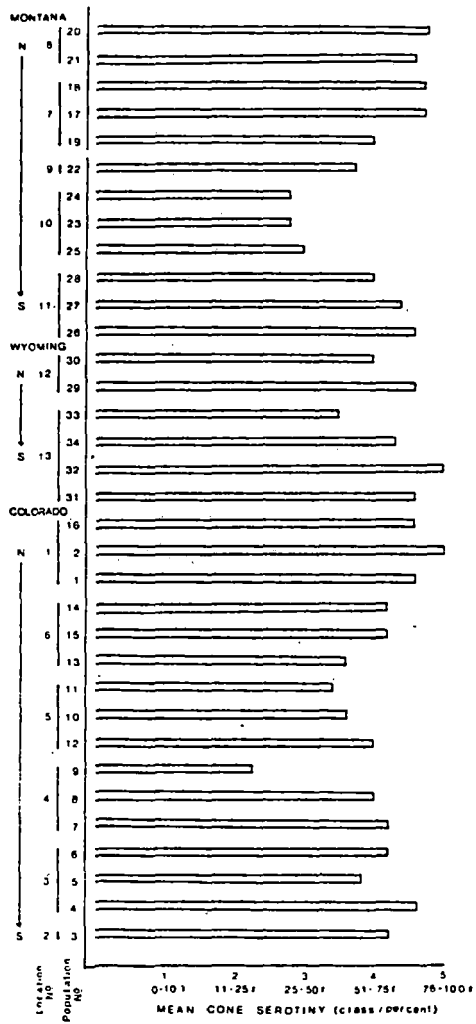


Table 13. Duncan's multiple range test on cone serotiny by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (class value)
9	2.3
23	2.8
24	2.8
25	3.0
11	3.4
33	3.5
10	3.6
13	3.6
22	3.8
5	3.8
8	4.0
12	4.0
19	4.0
28	4.0
30	4.0
3	4.2
6	4.2
7	4.2
14	4.2
15	4.2
34	4.3
27	4.4
1	4.6
4	4.6
16	4.6
21	4.6
26	4.6
29	4.6
31	4.6
17	4.8
18	4.8
20	4.8
2	5.0
32	5.0

Table 14. Duncan's multiple range test on cone serotiny by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (class value)
10	2.8
4	3.7
5	3.7
9	3.8
6	4.0
2	4.2
3	4.2
12	4.3
11	4.4
7	4.5
13	4.5
8	4.7
1	4.7

Figure 12. Geographic pattern of statistically different groups of sample locations based on cone serotiny. Explanation of the symbols is on page 33.

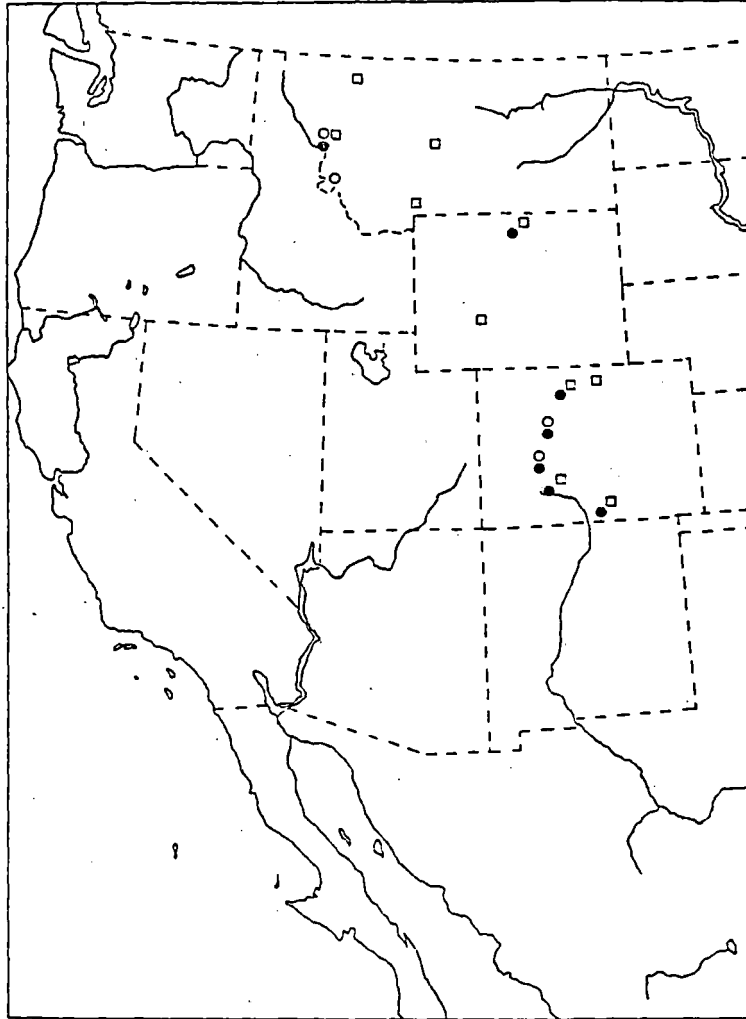


Figure 13. Cone abundance of sampled populations. Histograms correspond to population mean values.

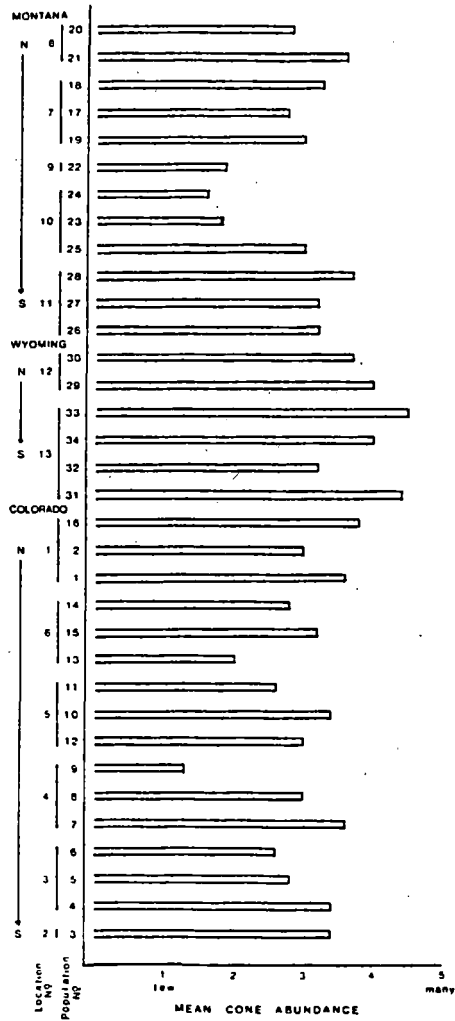


Table 15. Duncan's multiple range test on cone abundance by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (class value)
9	1.3
24	1.6
22	1.8
23	1.8
13	2.0
6	2.6
11	2.6
27	2.6
17	2.8
5	2.8
14	2.8
20	2.8
2	3.0
8	3.0
12	3.0
19	3.0
25	3.0
15	3.2
26	3.2
32	3.2
18	3.3
3	3.4
4	3.4
10	3.4
30	3.4
1	3.6
7	3.6
21	3.6
28	3.7
16	3.8
29	4.0
34	4.0
31	4.4
33	4.5

Table 16. Duncan's multiple range test on cone abundance by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (class value)
9	1.8
10	1.9
6	2.7
4	2.8
3	2.9
5	3.0
7	7.0
11	3.1
8	3.2
2	3.4
1	3.5
12	3.7
13	3.9

Figure 14 shows again the clinal nature of variation for cone abundance among latitudinal locations. Lowest cone abundance seemed to be in southwest Montana and highest abundance in Wyoming.

Since lodgepole pine does not have "mast years", when periodic, prolific cone production is greater than normal (Smith, 1970), the differences in cone abundance among the populations was probably not due to seasonal differences. Differing predation at the various collection areas may have accounted for some of the variation, but this factor was not specifically measured in this study. The pine squirrel (Tamiasciurus) has been shown to be an effective predator on lodgepole pine cones, and the evolutionary significance of the effect of pine squirrel predation on lodgepole pine cone morphology has been studied (Elliott, 1974).

Needle Persistence

Over all sample trees, needle persistence averaged between eight and nine years. Highly significant correlations were found between the number of years needles stayed on the tree and latitude, longitude, and elevation of the sample sites. Over all trees sampled, persistence of the needles was less at lower elevations, less in more northern sample sites, and less in trees from the more eastern parts of the sampled range.

Figure 15 shows that, on a population basis, needle persistence was for a greater number of years in more populations in the southern part of the sampled range than in populations farther to the north. Population-to-population variation was greater in the north for this trait, and this is also shown in Table 17.

Figure 14. Geographic pattern of statistically different groups of sample locations based on cone abundance. Explanation of the symbols is on page 33.

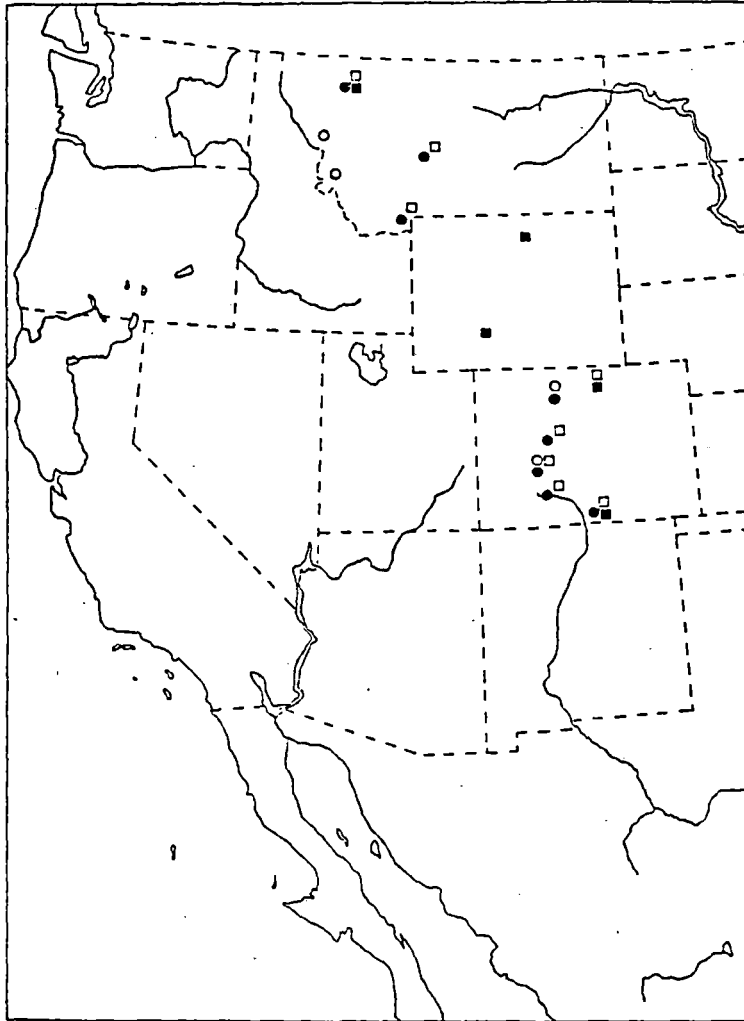


Figure 15. Needle persistence for sampled populations. Histograms correspond to population mean values.

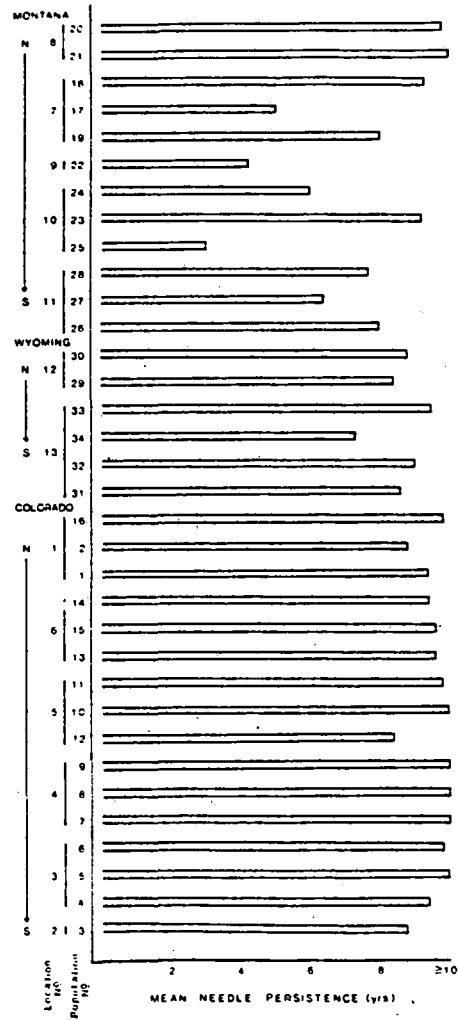


Table 17. Duncan's multiple range test on needle persistence by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (years)
25	3.0
22	4.3
17	5.0
24	6.0
27	6.4
34	7.3
28	7.7
19	8.0
26	8.0
12	8.4
29	8.4
31	8.6
2	8.8
3	8.8
30	8.8
32	9.0
23	9.2
18	9.3
1	9.4
4	9.4
14	9.4
33	9.5
13	9.6
15	9.6
6	9.8
11	9.8
16	9.8
20	9.8
5	10.0
7	10.0
8	10.0
9	10.0
10	10.0
21	10.0

Table 18 shows that one latitudinal location was distinctly different from all the rest for needle persistence. The Lolo Pass, Montana area had the lowest values for needle persistence of all sampled locations. Figure 16 illustrates the higher values for needle persistence in Colorado, but a pattern similar to the southern collection was also found near Marias Pass, Montana.

Results of Laboratory Measured Traits

Cone Length

Cone length varied from 29.4 to 50.2 millimeters of all trees sampled with a mean value of 37.7 millimeters. Cone length was significantly correlated with latitude, longitude, and elevation. Shorter cones were correlated significantly with more northern collections and collections from higher altitudes, and also significantly correlated with more western collection sites.

Figure 17 indicates that variation between mean values of cone length for populations was not great over the sampled geographic range. Table 19 also shows this, and shows the overlapping nature of the groups of population mean values. Similarly, when populations were grouped into latitudinal locations, the continuous nature of geographic variation was evident (Table 20). The shortest cones were from southwest Montana, and longer cones were found in the southern collections (Figure 18).

Considering all trees together, longitude accounted for a significant amount of the variation in cone length in the overall regression (Table 21). Topography, tree age, and tree diameter were

Table 18. Duncan's multiple range test on needle persistence by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (years)
9	4.31
10	6.8
11	7.3
7	7.4
12	8.6
13	8.6
2	8.8
1	9.3
5	9.4
6	9.5
8	9.9
4	10.0

Figure 16. Geographic pattern of statistically different groups of sample locations based on needle persistence. Explanation of the symbols is on page 33.

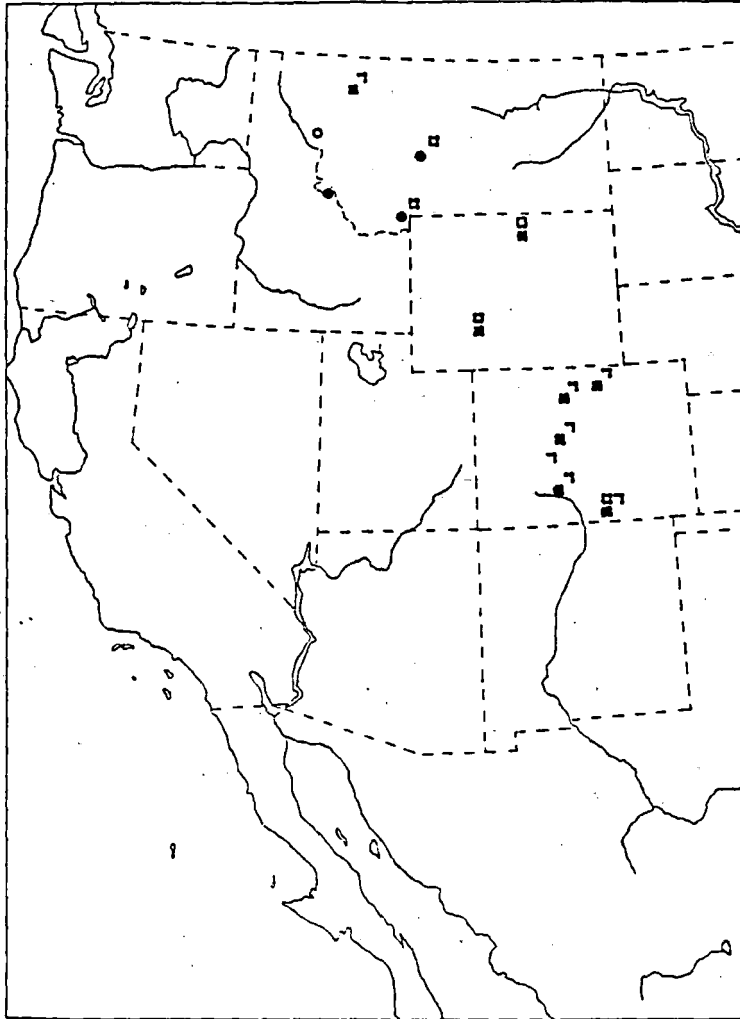


Figure 17. Cone length for sampled populations. Explanation of the symbols is on pages 31 - 32.

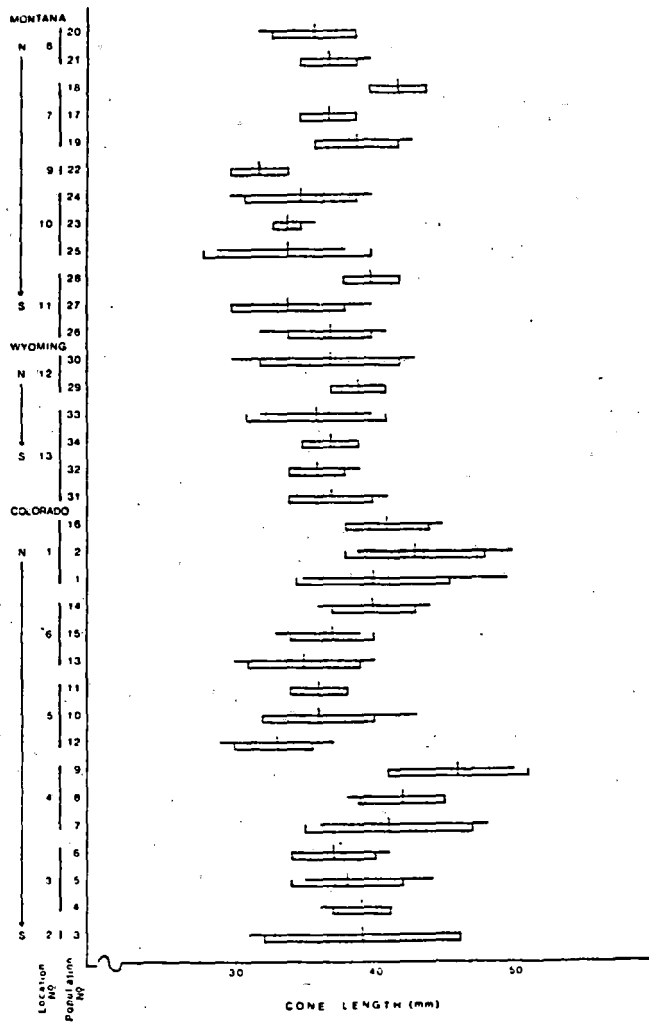


Table 19. Duncan's multiple range test on cone length by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (mm)
22	31.7
12	32.8
25	33.7
27	33.8
23	34.2
24	35.0
13	35.4
11	35.6
32	36.0
10	36.1
33	36.2
20	36.4
17	36.5
34	36.6
15	36.8
30	36.9
31	37.1
26	37.2
21	37.2
6	37.5
5	38.4
19	38.7
4	39.1
3	39.3
29	39.3
14	39.6
1	40.4
28	40.4
16	40.9
7	40.9
18	41.6
8	42.2
2	43.2
9	45.6

Table 20. Duncan's multiple range test on cone length by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (mm)
9	31.7
10	34.5
5	34.7
13	36.5
11	36.6
8	36.8
6	37.2
12	38.1
3	38.3
7	39.0
2	39.3
1	41.5
4	42.5

Figure 18. Geographic pattern of statistically different groups of sample locations based on cone length. Explanation of the symbols is on page 33.

Table 21. Regression analysis on cone length

Basis	Equation	R ²	F	P
Overall	87.11 - 0.46(long)	.11	(1,151): 17.914	< 0.001
Location 1 (CO-1, CO-2, CO-16)	No significant regression			
Location 3 (CO-4, CO-5, CO-6)	No significant regression			
Location 5 (CO-10, CO-11, CO-12)	25.76 + 3.36(topo)	.25	(1,13): 4.382	0.056
Location 7 (MT-1, MT-2, MT-3)	45.58 - 4.01(topo) - 0.03(age)	.56	(2,9): 5.68	0.025
Location 8 (MT-4, MT-5)	41.14 - 0.01(dia)	.29	(1,8): 3.248	0.109
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

significant in individual location regressions, but in three locations, no significant regression model could be developed.

Cone Width

Cone width ranged from 25.4 to 50.8 millimeters and averaged 35.9 millimeters over all trees sampled. There were no significant correlations, or even large correlation coefficients between cone width and any of the sample site characteristics. Population-to-population variation in cone width was not great and is illustrated in Figure 19 and Table 22. Additionally, the populations with the widest cones and least wide cones were adjacent (populations 9 and 12 in Colorado).

Greatly overlapping groups of statistically homogeneous populations and locations are illustrated in Table 22 and 23. No overall pattern was apparent for geographic variation in this trait on the basis of mean values for sample latitudinal locations (Figure 20).

Regression analysis, summarized in Table 24, showed that the geographic variables included in this study had little to do with explaining the variation in this trait.

Cone Shape

Cone shape, the ratio of cone length to width, varied from 0.9 to 1.4 for this study. Mean cone shape was 1.1, meaning that the average lodgepole pine cone was just a little longer than wide in this study when the cone was open.

Figure 19. Cone width for sampled populations. Explanation of the symbols is on pages 31 - 32.

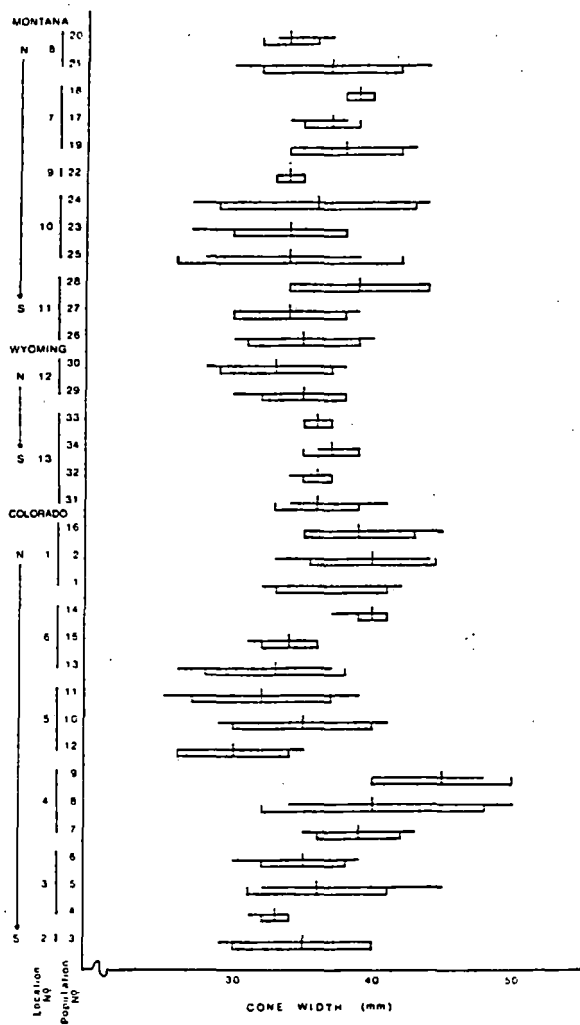


Table 22. Duncan's multiple range test on cone width by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (mm)
12	30.5
11	32.1
13	32.9
4	33.0
30	33.1
25	33.5
23	33.6
15	33.7
22	33.8
27	34.1
20	34.4
29	34.6
10	34.6
3	34.8
6	35.1
26	35.4
32	35.5
33	35.8
5	36.1
24	36.1
31	36.3
17	36.5
34	36.9
1	37.8
21	37.1
19	37.9
18	38.6
28	38.7
7	39.1
16	39.4
14	39.6
2	39.6
8	40.1
9	44.9

Table 23. Duncan's multiple range test on cone width by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (mm)
5	32.4
9	33.8
12	33.8
10	34.6
3	34.7
2	34.8
6	35.4
11	35.7
8	35.7
13	36.1
7	37.7
1	38.7
4	40.9

Figure 20. Geographic pattern of statistically different groups of sample locations based on cone width. Explanation of the symbols is on page 33.

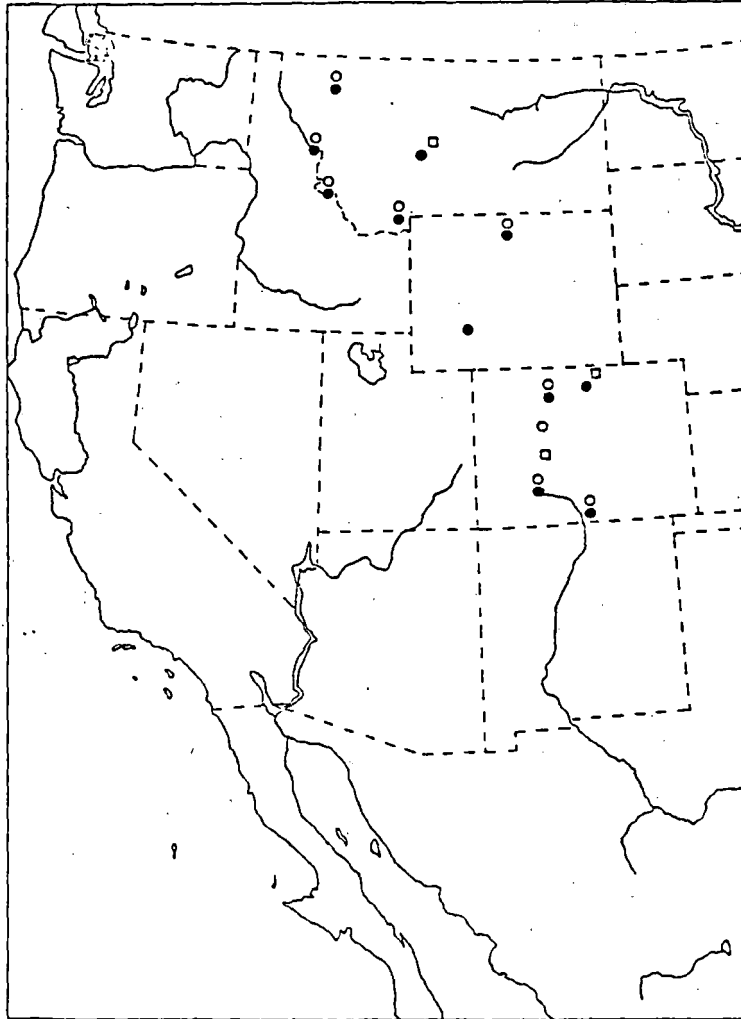


Table 24. Regression analysis on cone width.

Basis	Equation	R ²	F	P
Overall	No significant regression			
Location 1 (CO-1, CO-2, CO-16)	$45.43 - 0.77(ht) + 0.02(age)$	.27	(2,12): 2.534	0.121
Location 3 (CO-4, CO-5, CO-6)	No significant regression			
Location 5 (CO-10, CO-11, CO-12)	$55.80 - 1.35(ht) - 0.55(slope)$	.51	(2,12): 6.34	0.013
Location 7 (MT-1, MT-2, MT-3)	No significant regression			
Location 8 (MT-4, MT-5)	No significant regression			
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

Correlation analysis on cone shape paralleled that of cone length, and showed highly significant negative correlations of shape with latitude and longitude, and a highly significant positive correlation with elevation. Cones were relatively longer and thinner to the south and east, and shorter and fatter at lower elevations.

Figure 21 shows a slight trend for population mean values of cone shape to get smaller from south to north. Again, populations tended to have shorter and fatter cones in the north. Variation within populations for this trait was relatively uniform, population-to-population. Population 2 was more variable than the others, though.

Although among-population and among-location variation was significant, as illustrated by Tables 25 and 26, the groups of statistically different populations or locations overlapped broadly, and, for latitudinal locations, this broad overlap is geographically represented in Figure 22.

As was true for cone length, the only important independent variable in the overall regression model on cone shape was longitude. In the individual location regressions, slope of the sample site was important in locations 3 and 5, but tree age and tree height entered into the regression models in locations 5 and 8 (Table 27).

Cone Volume

Cone volume varied considerably between sample trees, ranging from 3.2 cc to 17.4 cc. Overall, average cone volume was 8.6cc. Cone volume was significantly correlated with latitude, longitude, and elevation. A significant correlation coefficient also occurred between cone volume and topsoil texture. In general, cone volume

Figure 21. Cone shape for sampled populations. Explanation of the symbols is on pages 31 - 32.

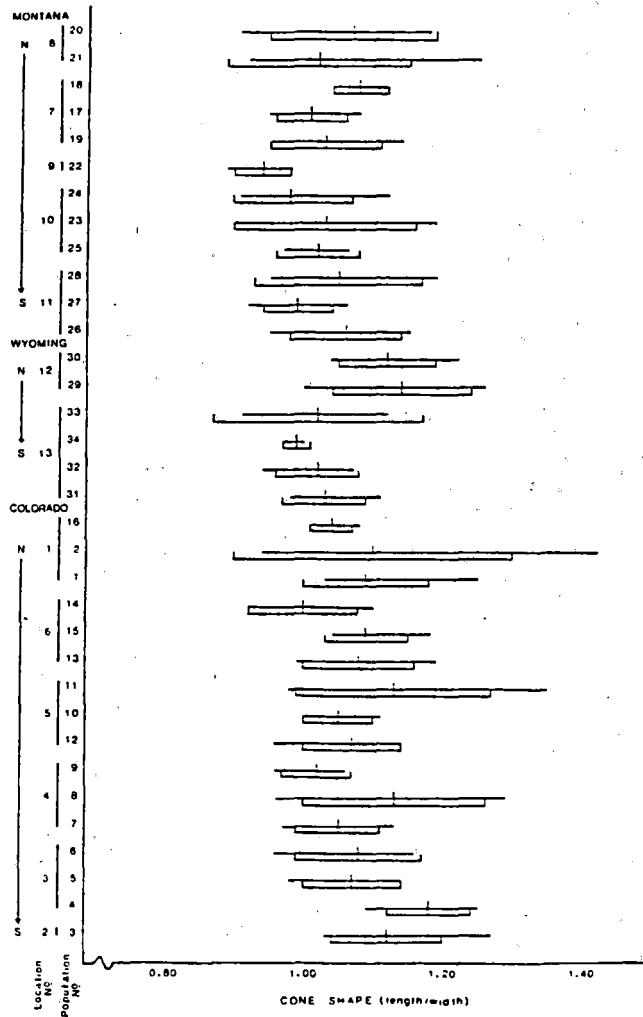


Table 25. Duncan's multiple range test on cone shape. by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (ratio)
22	0.94
24	0.98
34	0.99
27	1.00
14	1.01
17	1.01
25	1.02
33	1.02
9	1.02
21	1.02
32	1.02
31	1.03
19	1.03
23	1.03
16	1.04
7	1.05
28	1.05
10	1.05
26	1.06
20	1.07
5	1.07
12	1.07
6	1.08
18	1.08
13	1.08
15	1.09
1	1.09
2	1.11
30	1.12
3	1.12
11	1.13
8	1.13
29	1.14
4	1.18

Table 26. Duncan's multiple range test on cone shape by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (ratio)
9	0.94
10	1.01
13	1.01
11	1.03
7	1.04
8	1.04
6	1.06
4	1.07
1	1.08
5	1.08
3	1.11
2	1.12
12	1.13

Figure 22. Geographic pattern of statistically different groups of sample locations based on cone shape. Explanation of the symbols is on page 33.

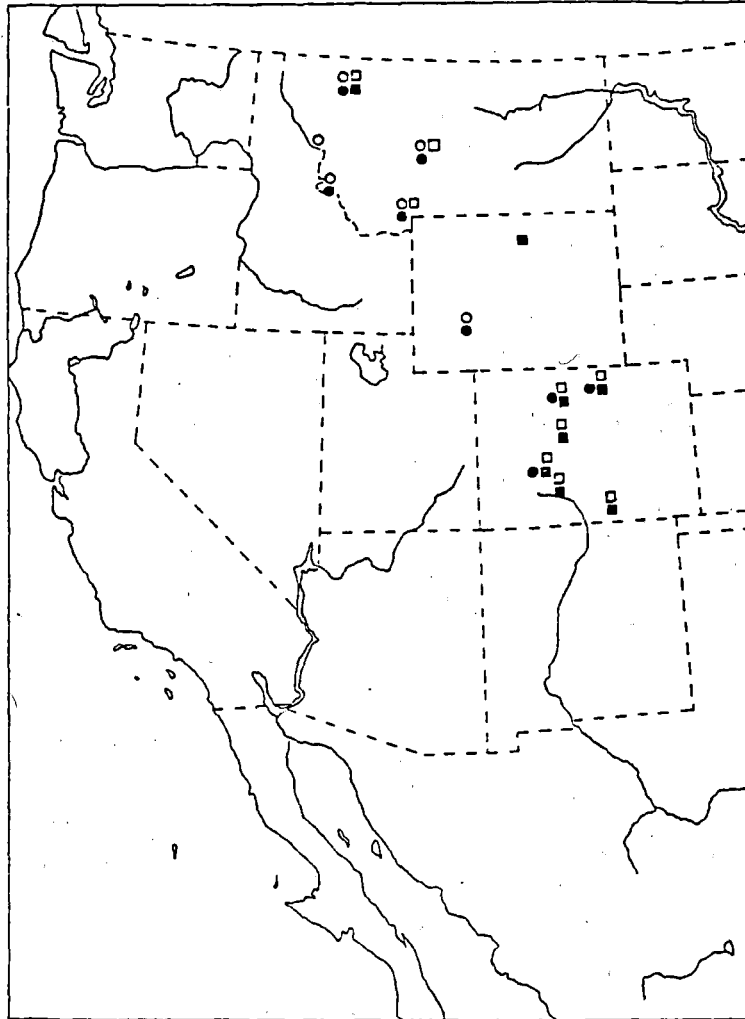


Table 27. Regression analysis on cone shape.

Basis	Equation	R ²	F	P
Overall	$2.20 - 0.01(\text{long})$	.11	(1,151): 19.479	< 0.001
Location 1 (CO-1, CO-2, CO-16)	No significant regression			
Location 3 (CO-4, CO-5, CO-6)	$1.35 - 0.006(\text{slope})$	.35	(1,13): 7.123	0.019
Location 5 (CO-10, CO-11, CO-12)	$0.73 + 0.002(\text{age}) + 0.008(\text{slope})$	.59	(2,12): 8.227	0.006
Location 7 (MT-1, MT-2, MT-3)	No significant regression			
Location 8 (MT-4, MT-5)	$0.65 - 0.003(\text{dia}) - 0.083(\text{ht})$	.54	(2,7): 4.162	0.064
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

increased with a decrease in latitude or a decrease in longitude. Greater cone volumes were found at higher elevations and at sites with more clay-like topsoil textures.

Figure 23 shows that within-population variability was fairly high for most populations for cone volume. Figure 23 illustrates a geographic grouping of populations: 1) populations 3 through 9 in southern Colorado, 2) populations 12 north through 16 in central and northern Colorado, 3) populations 31 north through 22 in Wyoming and southwestern Montana, and 4) populations 19 north through 20 in northern and central Montana. In groups 1) and 2) above, the trend was for increasing volume from the south to the north. In groups 3) and 4), the opposite trend was apparent.

Statistical grouping of the populations, shown in Table 28, again illustrates the continuous variation pattern typical for the traits measured in this study. Most populations were members of statistical groups which had component population mean values below 10.0 cc. Two groups of populations, each with only seven members, contained mean values from 10.2 to 13.2 cc.

On a latitudinal location basis, variation in cone volume was continuous, also (Table 29), and an overall trend towards larger cone volumes in southern sample locations is apparent in Figure 24.

Regression analysis, summarized in Table 30, showed that over all trees sampled, a model including longitude and latitude as independent variables was highly significant in explaining the variation observed in cone volume measures. At individual latitudinal locations, tree height and tree age often appeared as significant components of regression models.

Figure 23. Cone volume for sampled populations. Explanation of the symbols is on pages 31 - 32.

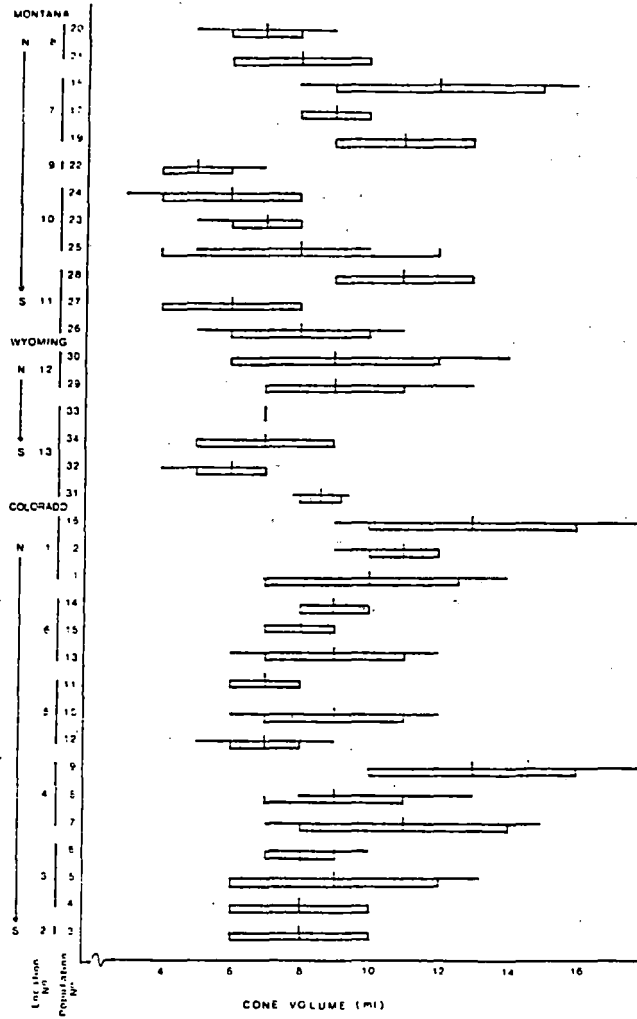


Table 28. Duncan's multiple range test on cone volume by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (ml)
22	5.4
24	5.8
32	6.1
27	6.4
23	6.6
12	6.7
20	6.7
11	6.9
33	7.2
34	7.3
25	7.5
21	7.6
15	7.8
4	8.4
26	8.4
3	8.4
6	8.4
10	8.6
31	8.7
17	8.7
13	8.7
5	8.9
29	9.0
14	9.1
30	9.4
8	9.4
1	10.2
28	10.9
19	10.9
2	11.2
7	11.2
18	12.1
9	13.2
16	13.2

Table 29. Duncan's multiple range test on cone volume by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (ml)
9	5.4
10	6.4
8	7.2
13	7.4
5	7.4
11	8.2
2	8.4
6	8.5
3	8.6
12	9.2
7	10.6
4	11.1
1	11.5

Figure 24. Geographic pattern of statistically different groups of sample locations based on cone volume. Explanation of the symbols is on page 33.

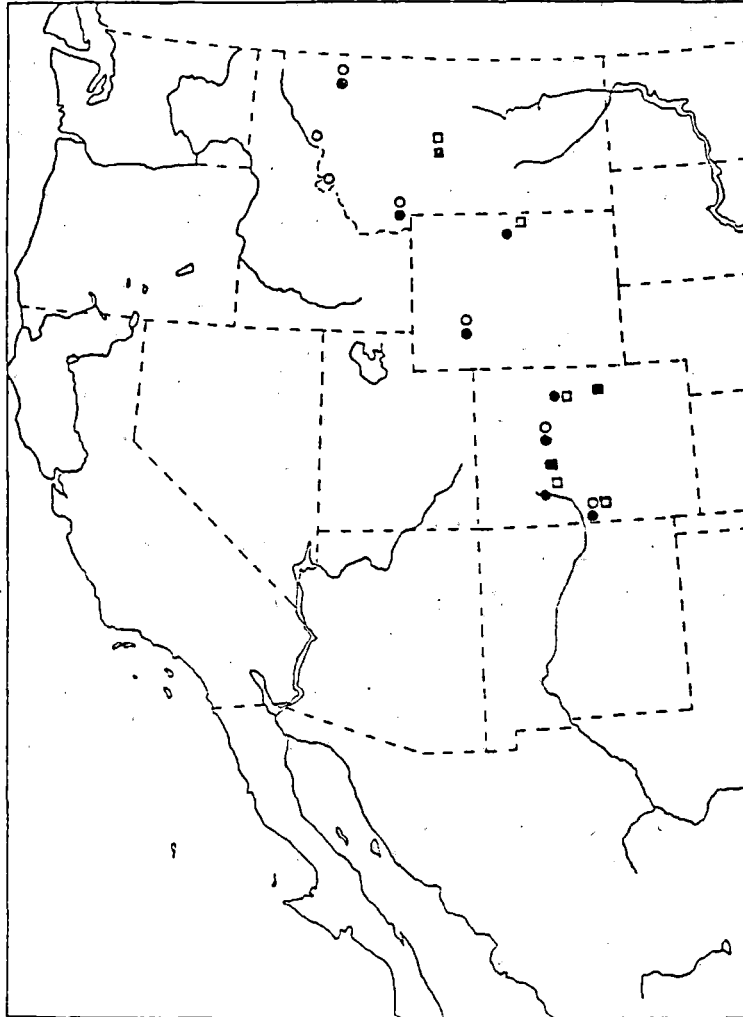


Table 30. Regression analysis on cone volume.

Basis	Equation	R ²	F	P
Overall	$59.95 - 0.61(\text{long}) + 0.34(\text{lat})$	.16	(2,150): 14.893	< 0.001
Location 1 (CO-1, CO-1, CO-16)	$13.52 + 0.02(\text{age}) - 0.41(\text{ht})$	.44	(2,12): 4.748	0.030
Location 3 (CO-4, CO-5, CO-6)	No significant regression			
Location 5 (CO-10, CO-11, CO-12)	$10.21 - 0.19(\text{slope})$	.21	(1,13): 3.554	0.082
Location 7 (MT-1, MT-2, MT-3)	$11.97 - 2.35(\text{topo}) + 0.02(\text{dia})$	.46	(2,9): 3.772	0.065
Location 8 (MT-4, MT-5)	$15.33 - 0.67(\text{ht})$	.40	(1,8): 5.227	0.052
Location 13 (WY-3, WY-4, WY-5 WY-5A)	$11.60 - 0.99(\text{soil}) - 0.01(\text{age})$	.44	(2,12): 4.758	0.023

Apophysis Shape

Apophysis shape was evaluated as a class-type variable which varied between flat (class 1) and attenuate (class 5). Over all trees sampled, an average value for apophysis shape was 3.4. To significant correlations were found between apophysis shape and any of the sample site variables.

Figure 25 shows no great difference between any of the populations for apophysis shape, and no geographic trends by population were clear. Duncan's multiple range test gave two, large statistical groupings of the populations which overlapped to a great extent (Table 31). Similar results were revealed for location-to-location variation patterns (Table 32) and the geographic distribution of statistically different groups of locations showed no clear pattern (Figure 26).

Cone Weight

The average, dry, open-cone weight of a lodgepole pine cone sampled in this study varied from 1.8 to 9.9 grams, and averaged 5.6 grams. No significant correlations were found, however, between cone weight and any of the measured site characteristics.

Figure 27 shows that the variation within populations was much the same for all populations. Although no significant correlations were found between cone weight and any of the geographic variables, patterns for cone weight (Figure 27) are similar to those for cone volume (Figure 23). The four geographic groups suggested for cone volume seem appropriate for cone weight, as well. There was an overall increase in cone weight from latitudinal location 2 to

Figure 25. Apophysis shape for sampled populations. Histograms correspond to population mean values.

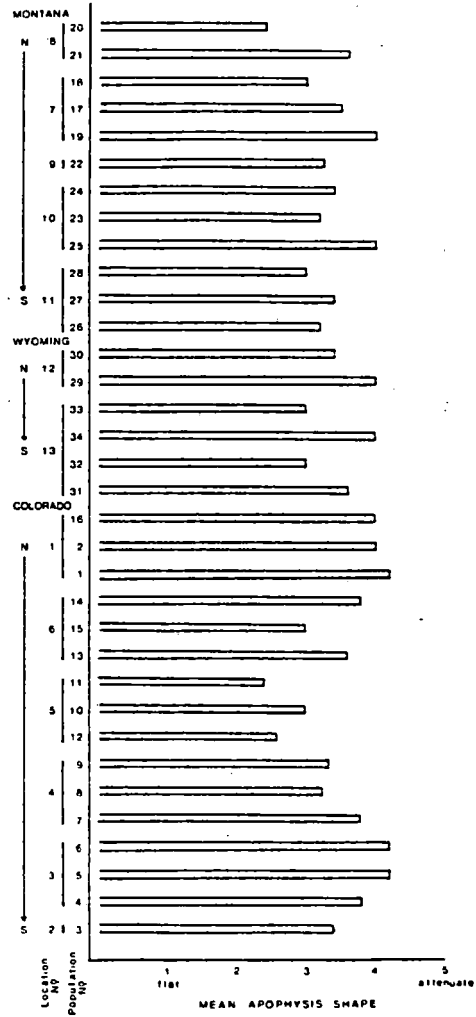


Table 31. Duncan's multiple range test on apophysis shape by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (class value)
11	2.4
20	2.4
12	2.6
10	3.0
15	3.0
18	3.0
28	3.0
32	3.0
33	3.0
23	3.2
26	3.2
8	3.3
22	3.3
9	3.3
3	3.4
24	3.4
27	3.4
30	3.4
17	3.5
4	3.6
13	3.6
21	3.6
31	3.6
5	3.8
7	3.8
14	3.8
2	4.0
16	4.0
19	4.0
25	4.0
29	4.0
34	4.0
1	4.2
6	4.2

Table 32. Duncan's multiple range test on apophysis shape by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (class value)
5	2.7
8	3.0
11	3.2
9	3.3
2	3.4
13	3.4
10	3.4
6	3.5
4	3.5
7	3.5
12	3.7
3	3.9
1	4.1

Figure 26. Geographic pattern of statistically different groups of sample locations based on apophysis shape. Explanation of the symbols is on page 33.

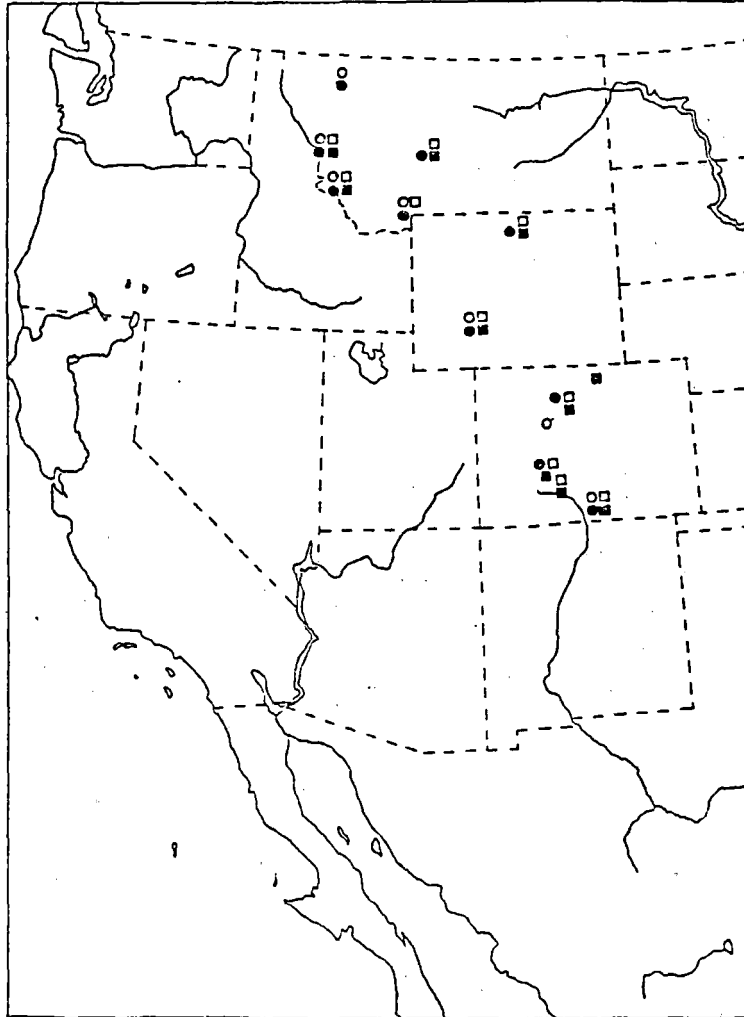
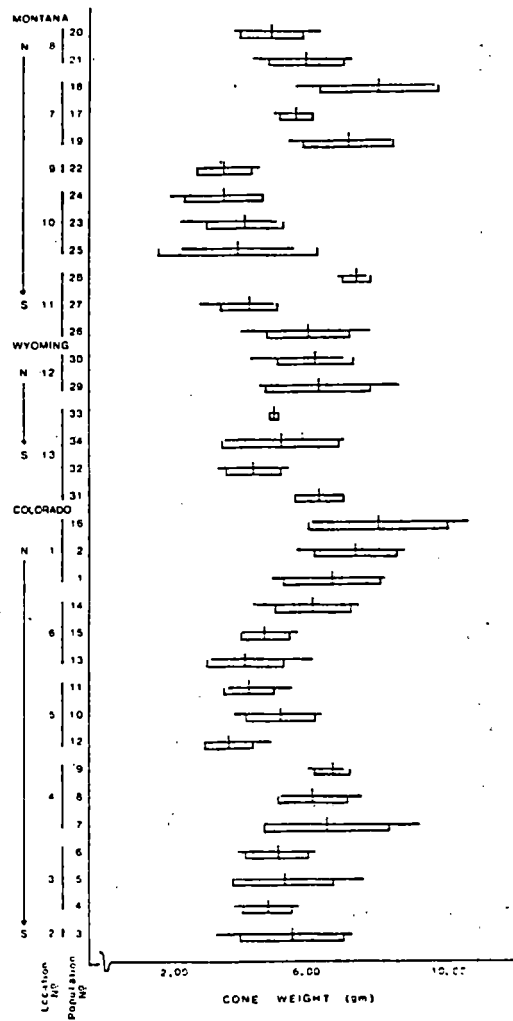


Figure 27. Cone weight of sampled populations. Explanation of the symbols is on pages 31 - 32.



location 4. Within locations 3 and 4, an increase in cone weight with an increase in elevation was shown. Similarly, from location 5 north to location 1 there was an increase in cone weight and the populations within all three locations showed an increase in cone weight with an increase in elevation. A third geographic group composed of locations 10, 11, 12, and 13 showed opposite trends as the two geographic groups just described above. Locations 7 and 8 form a group similar in character to the group with locations 10, 11, 12, and 13 as components.

Duncan's multiple range test, both on populations and on locations, indicated continuous variation in broadly overlapping, statistical groups (Table 33, Table 34). Figure 28 shows the clinal pattern of statistical grouping of latitudinal locations, and shows that sample locations in southwest Montana contained the lowest values for cone weight, high values were found throughout the sampled range, and the greatest variability (the location belonging to the greatest number of statistical groups) was in southern Colorado.

Overall regression analysis, summarized in Table 35, showed longitude and latitude as significantly important in explaining the variation in cone weight. At the individual latitudinal locations, sample tree diameter was important in location 3, slope and sample tree height were important in location 5, sample tree diameter, position on slope, and sample elevation were important at location 7, sample tree height and sample elevation were important at location 8, and topsoil texture and position on slope were important at location 13 for explaining the variation in cone weight.

Table 33. Duncan's multiple range test on cone weight by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (grams)
24	3.77
22	3.82
12	3.85
25	4.23
13	4.31
23	4.39
11	4.44
27	4.47
32	4.60
15	4.87
4	4.90
6	5.18
33	5.20
20	5.23
10	5.32
34	5.41
5	5.42
3	5.56
17	5.88
26	6.12
8	6.17
21	6.22
14	6.27
30	6.39
31	6.46
29	6.52
7	6.65
9	6.77
1	6.78
19	7.42
28	7.57
2	7.68
16	8.15
18	8.27

Table 34. Duncan's multiple range test on cone weight by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (grams)
9	3.82
10	4.11
5	4.54
6	5.15
3	5.17
13	5.46
2	5.56
8	5.72
11	5.82
12	6.45
4	6.52
1	6.87
7	7.18

Figure 28. Geographic pattern of statistically different groups of sample locations based on cone weight. Explanation of the symbols is on page 33.

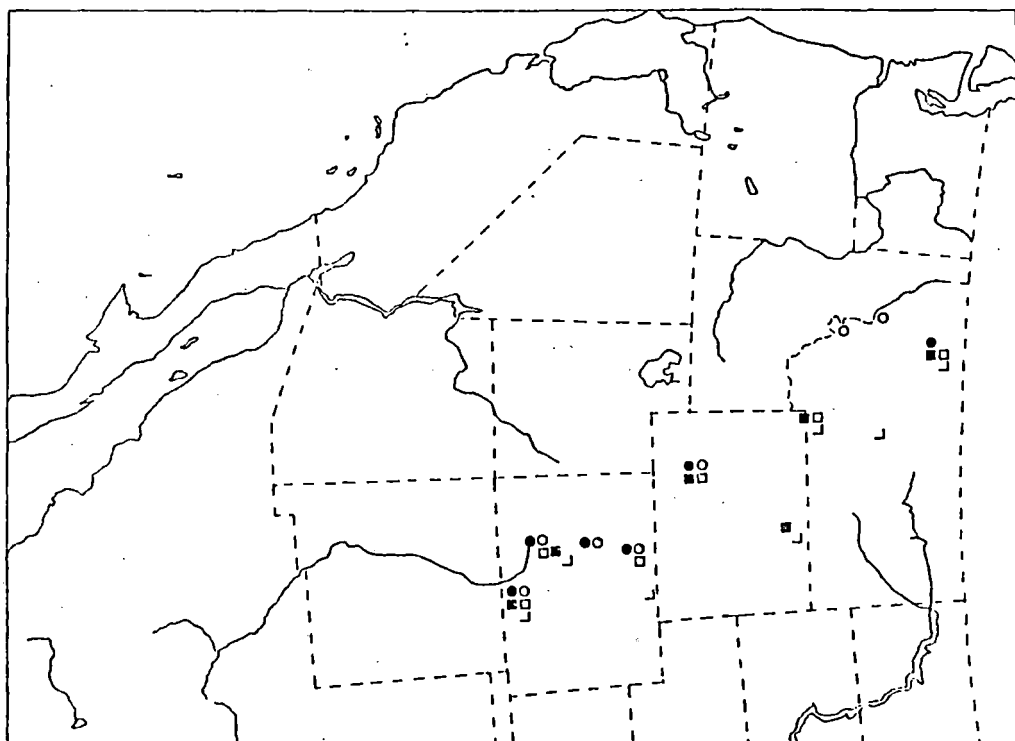


Table 35. Regression analysis on cone weight.

Basis	Equation	R ²	F	P
Overall	$33.72 - 0.40(\text{long}) + 0.35(\text{lat})$	.16	(2,150): 15.117	< 0.001
Location 1 (CO-1, CO-2, CO-16)	No significant regression			
Location 3 (CO-4, CO-5, CO-6)	$4.13 + 0.006(\text{dia})$	.13	(1,13): 2.023	0.179
Location 5 (CO-10, CO-11, CO-12)	$8.42 - 0.16(\text{slope}) - 0.13(\text{ht})$	.46	(2,12): 5.232	0.023
Location 7 (MT-1, MT-1, MT-3)	$4.88 - 1.69(\text{topo}) + 0.01(\text{dia}) + 0.002(\text{elev})$	.62	(3,8): 4.393	0.042
Location 8 (MT-4, MT-5)	$60.35 - 0.44(\text{ht}) - 0.03(\text{elev})$	.59	(2,7): 4.984	0.045
Location 13 (WY-3, WY-4, WY-5 WY-5A)	$8.64 - 0.46(\text{soil}) - 0.63(\text{topo})$	.40	(2,12): 4.056	0.045

Cone Specific Gravity

Cone specific gravity averaged 67.1 over all trees sampled, but ranged from 42.0 to 89.0. Highly significant correlations were found for this trait with latitude, longitude, elevation, and topsoil texture. Cone specific gravity decreased from north to south, decreased from west to east, but increased from high elevation to low elevation, in general.

The gradual, distinct trend for population mean values of cone specific gravity to increase from the south part of the sampled range to the north is apparent as shown in Figure 29. Duncan's multiple range test on population mean values indicated that few populations were statistically grouped at the lower values for cone specific gravity, and most of those populations were from Colorado. Most statistically different groups of populations contained a large number of populations, and the continuous nature of the variation in this trait was apparent by the broad overlap of those groups. Similarly, statistical grouping of latitudinal locations shown in Table 37 indicates continuous variation. Figure 30 graphically shows the trend for increasing specific gravity with increasing latitude.

Latitude alone accounted for 23 percent of the variation in cone specific gravity in the overall regression equation (Table 38). Significant regressions for individual locations were developed for locations 1 and 5, only. At location 1, slope percent was the only significant independent variable, and at location 5, aspect was the only significant independent variable.

Figure 29. Cone specific gravity for sampled populations. Explanation of the symbols is on pages 31 - 32.

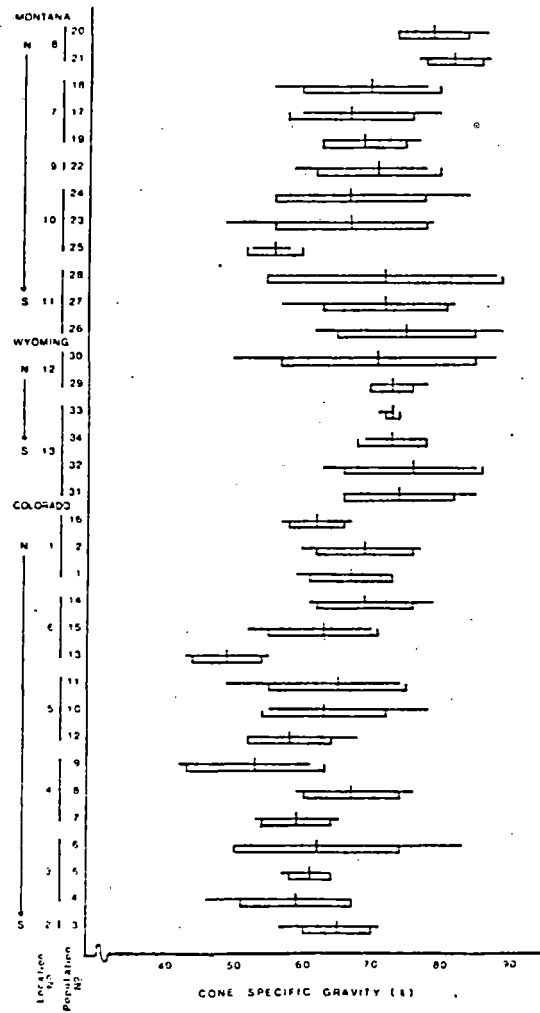


Table 36. Duncan's multiple range test on cone specific gravity by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (percent)
13	49.4
9	52.7
25	55.5
12	58.4
4	59.0
7	59.4
5	60.8
16	61.6
6	62.2
15	62.6
10	63.0
3	65.0
11	65.0
8	66.5
23	66.6
24	67.2
1	67.4
17	67.7
19	68.5
2	68.6
14	69.0
18	70.0
22	70.8
30	71.0
28	71.7
27	72.0
33	72.5
29	73.0
34	73.3
31	74.4
26	75.0
32	76.0
20	79.0
21	81.8

Table 37. Duncan's multiple range test on cone specific gravity by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (percent)
4	60.1
6	60.3
3	60.7
5	62.1
2	65.0
10	65.0
1	65.9
7	68.8
9	70.8
12	72.0
11	73.1
13	74.5
8	80.4

Figure 30. Geographic pattern of statistically different groups of sample locations based on cone specific gravity. Explanation of the symbols is on page 33.

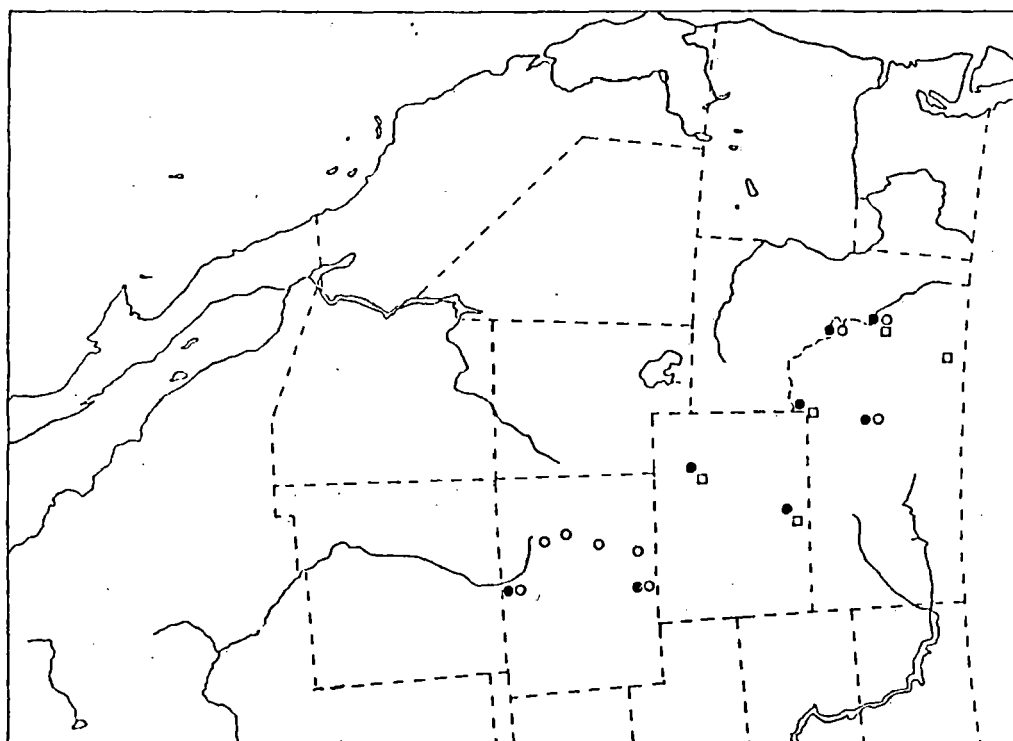


Table 38. Regression analysis on cone specific gravity.

Basis	Equation	R ²	F	P
Overall	$4.38 - 1.48(\text{lat})$	.23	(1,151): 44.127	< 0.001
Location 1 (CO-1, CO-2, CO-16)	$61.60 + 0.64(\text{slope})$	.26	(1,13): 4.608	0.051
Location 3 (CO-4, CO-5, CO-6)	No significant regression			
Location 5 (CO-10, CO-11, CO-12)	$51.85 + 1.62(\text{aspect})$	.12	(1,13): 1.785	0.205
Location 7 (MT-1, MT-2, MT-3)	No significant regression			
Location 8 (MT-4, MT-5)	No significant regression			
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

Needle Biomass

The average oven-dry weight of a fascicle of needles from a lodgepole pine tree included in this study was 3.5 mg. Oven-dry weights ranged from 2.0 to 6.1 mg. Significant correlations were found between needle biomass and latitude, longitude, and topographic position on the slope.

Strong latitudinal or elevational trends were not apparent in Figure 31, and population-to-population variation was broadly continuous, as shown in Table 39. Samples were lacking from populations 1 and 2, but their inclusion would not be expected to radically change the appearance of Figure 31. Duncan's multiple range test for location-to-location variation patterns showed a discontinuity between location 1 and the remainder of the locations, but that break was probably due to the lack of sample material from populations 1 and 2 (Table 40). Otherwise, there was no difference between any of the location mean values for needle biomass, and no map was developed for this trait.

Regression analysis for needle biomass is summarized in Table 41. In the overall model, topographic position on slope, slope percent, and sample tree age contributed significantly to explaining the variation in needle biomass. Locations with significant regressions included location 7, where sample tree age and topographic position on slope were important; location 8, where sample site elevation as well as sample tree age, height, and diameter entered the model; and location 13, where slope percent, sample tree diameter, and sample tree height were significant.

Figure 31. Needle biomass of sampled populations. Explanation of the symbols is on pages 31 - 32.

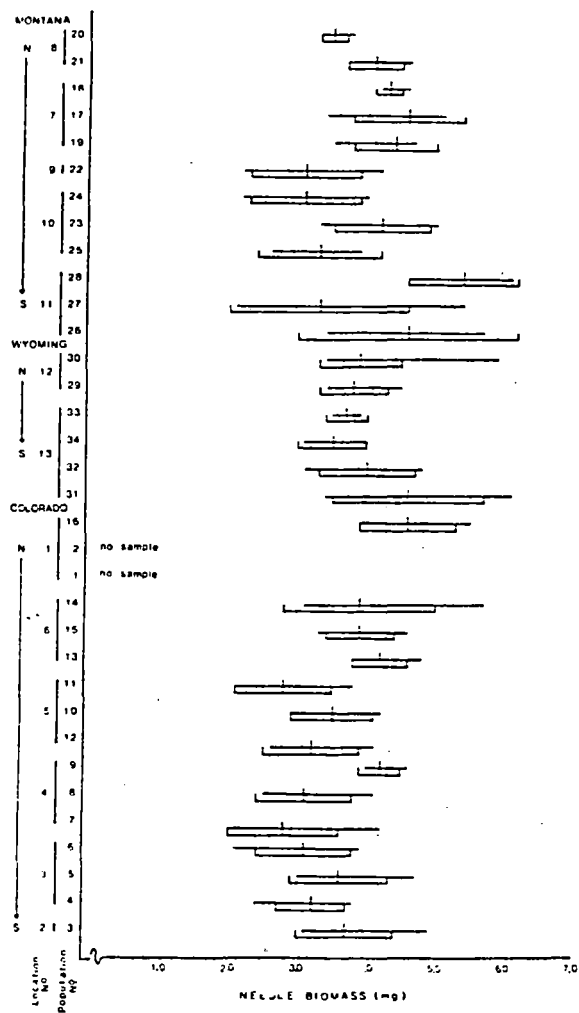


Table 39. Duncan's multiple range test on needle biomass by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (mg)
1	no sample
2	no sample
7	2.8
11	2.8
8	3.1
22	3.1
6	3.1
24	3.1
12	3.2
4	3.2
18	3.3
25	3.3
27	3.3
10	3.5
34	3.5
20	3.5
5	3.6
3	3.7
33	3.7
15	3.8
29	3.8
14	3.9
32	4.0
21	4.1
23	4.2
13	4.2
9	4.2
19	4.4
30	4.5
17	4.6
26	4.6
16	4.6
31	4.6
28	5.4

Table 40. Duncan's multiple range test on needle biomass by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (mg)
1	1.51
9	3.1
5	3.2
4	3.2
3	3.3
11	3.6
10	3.6
2	3.7
8	3.8
6	4.0
7	4.1
13	4.1
12	4.2

Table 41. Regression analysis on needle biomass.

Basis	Equation	R ²	F	P
Overall	$-.33 + .90(\text{topo}) + 0.03(\text{slope}) + 0.003(\text{age})$	.20	(3,149): 12.632	< .001
Location 1 (CO-1, CO-2, CO-16)	No significant regression			
Location 3 (CO-4, CO-5, CO-6)	No significant regression			
Location 5 (CO-10, CO-11, CO-12)	No significant regression			
Location 7 (MT-1, MT-2, MT-3)	$2.98 - 0.02(\text{age}) + 1.11(\text{topo})$	.41	(2,9): 3.147	0.092
Location 8 (MT-4, MT-5)	$32.88 - 0.02(\text{elev}) - 0.03(\text{age}) + 0.006(\text{dia}) - 0.11(\text{ht})$	.93	(4,5): 17.23	0.004
Location 13 (WY-3, WY-4, WY-5, WY-5A)	$3.66 + 0.20(\text{slope}) + 0.88(\text{dia}) - 0.47(\text{ht})$	.32	(3,11): 7.779	0.005

Results of Seedling Measurement

Greenhouse Seedling Height

Seedling growth under greenhouse conditions varied greatly among the populations and latitudinal locations. Average height of a lodgepole pine seedling after one "season" of greenhouse growth under accelerated growth conditions was 53.3 mm., and ranged from 23.0 to 91.6 mm. The only factor significantly correlated with greenhouse seedling height was elevation. Mean heights of seedlings from sample trees at higher elevations were lower than heights of seedlings from trees sampled at lower elevations.

Figure 32 shows the great variation within populations for this trait. A possible cause of the great within-population variability was the fact that some seedlings underwent two or three extra growth flushes under the artificial conditions of the greenhouse.

As may be expected, population-to-population variation for greenhouse seedling height growth was broadly continuous, as illustrated in Table 42. A similar pattern developed for location-to-location variation and is shown in Table 43. Broad, geographic overlap of mean values for latitudinal locations appear in Figure 33.

Overall regression of greenhouse seedling height on geographic variables was weak, yet elevation and latitude did show as significant in explaining the variation in this trait. Only locations 3, 5, and 7 showed significant regressions for greenhouse seedling height.

Figure 32. Greenhouse seedling height for sampled populations.
Explanation of the symbols is on pages 31 - 32.

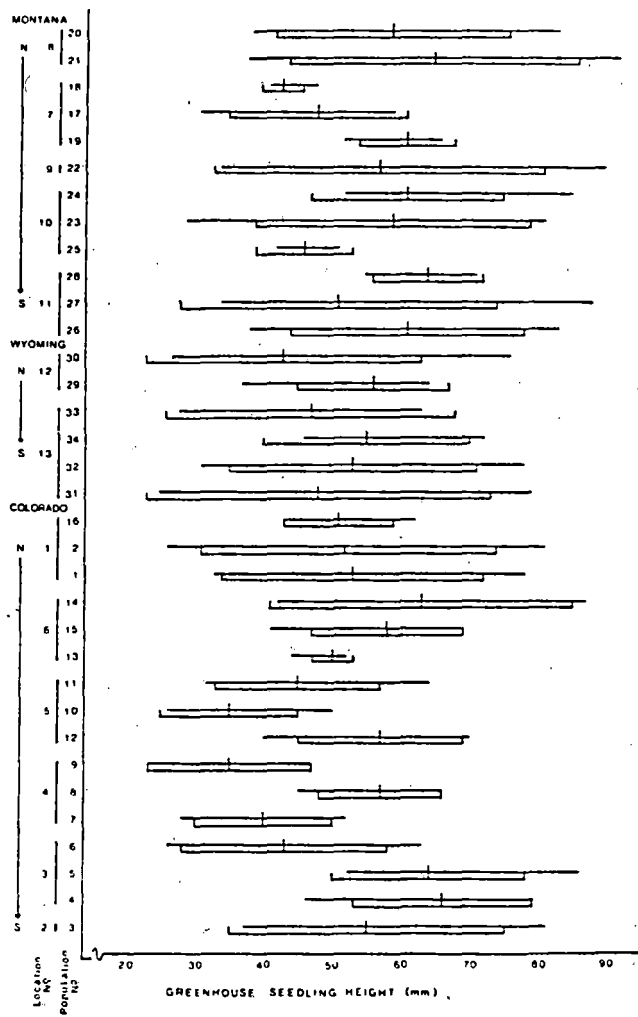


Table 42. Duncan's multiple range test on greenhouse height by population. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Population Number	Mean Value (mm)
10	35.1
9	35.4
7	39.7
6	43.0
18	43.0
30	43.4
11	44.7
25	46.2
33	47.5
17	47.6
31	48.3
13	49.8
16	50.7
27	51.5
2	52.2
1	52.7
32	53.5
34	55.4
3	55.4
29	56.2
22	56.7
8	56.8
12	57.4
15	57.6
20	58.8
23	59.3
24	60.6
26	61.2
19	61.5
14	62.9
28	63.7
5	63.9
21	64.8
4	65.7

Table 43. Duncan's multiple range test on greenhouse height by location. Mean values spanned by a vertical line are not significantly different at the 5 percent level.

Location Number	Mean Value (mm)
4	44.3
5	45.7
12	49.8
7	50.7
13	51.3
1	51.9
2	55.4
9	56.7
6	56.8
3	57.6
10	57.7
11	58.1
8	61.8

Figure 33. Geographic pattern of statistically different groups of sample locations based on greenhouse seedling height. Explanation of the symbols is on page 33.

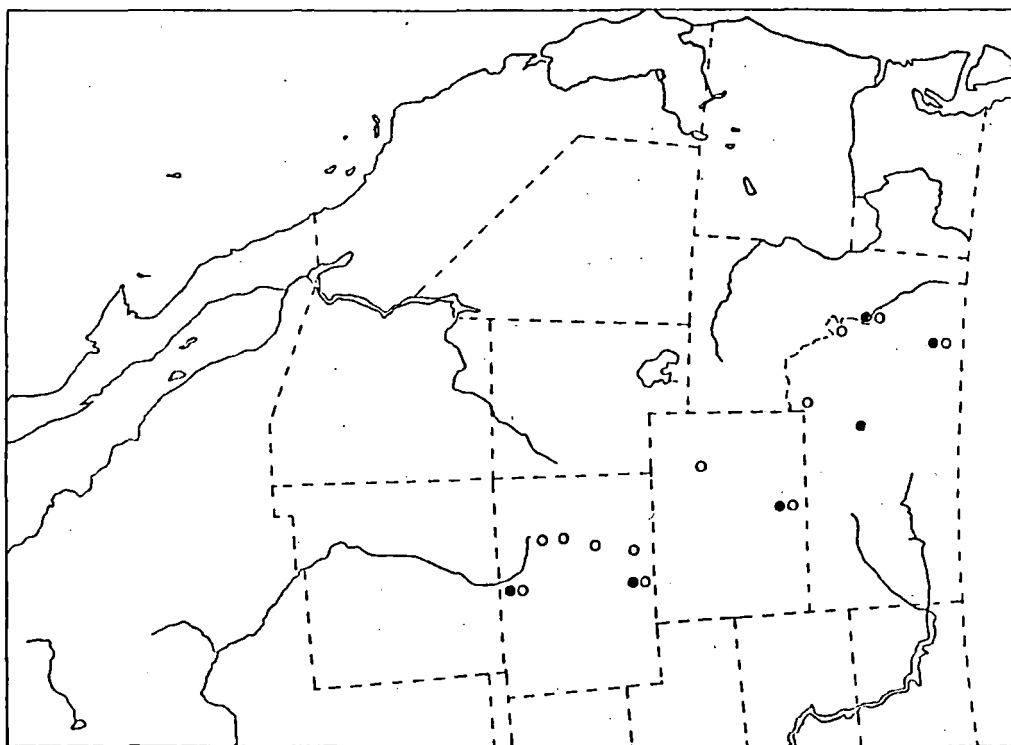


Table 44. Regression analysis on greenhouse seedling height.

Basis	Equation	R ²	F	P
Overall	$104.31 - 0.018(\text{elev}) - 2.04(\text{lat})$	.09	(2,150): 7.191	0.001
Location 1 (CO-1, CO-2, CO-16)	No significant regression			
Location 3 (CO-4, CO-5, CO-6)	$12.32 + 7.98(\text{aspect})$	.39	(1,13): 8.165	0.013
Location 5 (CO-10, CO-11, CO-12)	$12.33 + 2.23(\text{slope})$	.46	(1,13): 10.885	0.006
Location 7 (MT-1, MT-2, MT-3)	$73.62 - 0.63(\text{slope})$	.51	(1,10): 10.327	0.009
Location 8 (MT-4, MT-5)	No significant regression			
Location 13 (WY-3, WY-4, WY-5, WY-5A)	No significant regression			

Aspect was important at location 3 and slope percent was important at both location 5 and location 7 (Table 44).

Chapter IV. Discussion

Significant geographic variation among natural populations of lodgepole pine existed for every trait measured in this study. However, discernable trends in the variation patterns were not consistent, trait-to-trait. Several possibilities for interpreting the nature of phenotypic, geographic variation found in this study are discussed in this section.

Variation in Cone Traits

Those traits which are often considered to be under strong genetic control and also those which showed some of the more interesting variation patterns were related to the seed cones; particularly interesting were cone serotiny, cone specific gravity, and the manner in which cones were attached to the twigs.

Shaw (1914) listed several traits of seed cones in the genus Pinus which he considered to be indicators of evolutionary change. Shaw suggested that light weight, dehiscent, symmetrical cones were more "primitive" than dense, indehiscent, asymmetric forms. Coevolution of seed-eating mammals (such as the pine squirrel) with conifers has been suggested as a possible explanation for such evolutionary change, specifically in the development of the morphology of lodgepole pine cones (Elliott, 1973). Genera and species of the pine family

as well as rodents and placental herbivores began to appear about the same time: the late Paleocene and early Eocene (Smith, 1970). Smith suggested that as animals become more efficient at finding and extracting seeds, plants evolved so as to put more energy into seed-protective characteristics than into the production of more or bigger seeds. Elliott (1973) showed that squirrels have discriminative feeding behaviors which make them specific selective agents on cone morphology in lodgepole pine. Elliott's evidence indicated that the response of lodgepole pine to squirrel predation has been to develop more protective tissue in cone scales at the expense of number of seed per cone. Cone predation by squirrels would favor wider cones and a decreasing cone length to cone width ratio, according to Elliott. The present study did not investigate squirrel predation on cones in any of the populations sampled. However, the selective effects of squirrels or other cone predators could have partially caused some of the irregular patterns associated with cone morphology found in this study. High variability in traits such as apophysis shape, cone density, cone width, and cone orientation angle could have been reflecting a tree population adaptation, ensuring successful regeneration by at least some members of a given natural population of lodgepole pine in an area with an associated population of cone predators such as pine squirrels.

The heritability of the closed-cone habit (serotiny) has been studied in both lodgepole pine and the closely related jack pine (Teich, 1970; Lotan, 1976). Teich compiled data from several natural populations and hypothesized a one-gene, two-allele system controlling the closed-cone habit. He suggested that phenotypic expression of the

homozygous serotinous class was little affected by the environment, in the absence of severe heat, cones remained closed. Conversely, the homozygous non-serotinous class commonly showed open cones under normal stand temperatures. However, the heterozygous class could be easily affected by the temperature environment and cause inflated estimates of either homozygous class. Critchfield (1957, 1978) suggested that elevational clines existed for many traits of lodgepole pine, among them, cone serotiny. He presented evidence for an increase in open-cone habit with an increase in elevation. Illingworth (1970) also suggested elevational clines for serotiny in lodgepole pine in British Columbia. Lotan (1976) mentioned that the concept of an elevational cline for serotiny in the Rocky Mountains was an oversimplification of a much more complex pattern. Lotan suggested that fire was a strong selection force for serotiny and associated traits, and that frequent, intense fires favored the retention in a population of genes responsible for closed-cone habit. If Lotan's hypothesis has merit, then the distribution of the closed-cone habit should mirror the fire history of any given stand or population of lodgepole pine. Given the frequent but unpredictable occurrence of fire in the Rocky Mountains, now as well as in the distant past, it seems unlikely that a simple elevational cline exists for this trait. The present study gives evidence supporting the view that a simple, elevational cline does not exist for cone serotiny in the Rocky Mountains.

Critchfield has revealed patterns for cone specific gravity and associated traits (such as apophysis form) which parallel trends in cone serotiny. The present study did not show such clear parallels.

Although, over all trees sampled, cone density significantly decreased with an increase in elevation, no such trend was found with apophysis shape; all three traits - serotiny, cone specific gravity, and apophysis shape - appeared to have independent population-to-population variation patterns.

In a latitudinal context, however, the results of this study did agree with Critchfield's observation of increasing specific gravity with increasing latitude. Critchfield's mean values of 0.67 and 0.76 for the specific gravity of cones in his two Colorado collections are within the range of the present study. His mean values of 0.75 and 0.76 for two Montana collections are also within the present study range. It is difficult to compare the results of Critchfield's analysis with the present study because his sampling included only four natural populations within the geographic range examined here.

Influence of Jack Pine on Lodgepole Pine Cone Morphology

Jack pine is considered to be closely related to lodgepole pine, and studies comparing the two are common. Natural hybridization occurs where the two species ranges overlap in Canada, and fertile hybrids have been produced artificially (Moss, 1949; Mirov, 1956; Rehfeldt and Lotan, 1970; Richter and Stockwell, 1949).

Schoenike (1976) presented values of cone specific gravity for jack pine ranging from 0.64 to 1.06 with a mean of 0.92. Jack pine cones are evidently more dense than lodgepole pine cones. The higher values for density of lodgepole pine cones in the more northern collections of this study may indicate introgression of jack pine genes

for this trait into lodgepole pine. Evidence of a gradient of introgression of jack pine genes into lodgepole pine has been presented on the basis of chemical components of wood of the two species and putative hybrids (Zavarin et al., 1969; Mirov, 1956). A similar nothocline in the other direction, from lodgepole pine into jack pine, has not been well documented (Stern and Roche, 1974).

Cone orientation angle may show the strongest morphological evidence of jack pine influence on lodgepole pine (Critchfield, 1978). Cone angle is the angle formed by the relatively straight cone axis and the terminal part of the cone-bearing twig. Critchfield found cone angles ranging from 45° to 155° but averaging greater than 90° for lodgepole pine. Results of the present study were similar. For Rocky Mountain populations of lodgepole pine, cone angle ranged from 40° to 135° with a mean value of 106° . Cone angle in jack pine has been shown to range from -70° (curved below the branch, opposite the point of attachment to the twig) to 149° with a mean value of 28° . In the present study, cone angle significantly decreased south to north for lodgepole pine. Additionally, variation within populations was higher for cone angle in more northern samples as compared to southern samples.

Critchfield (1980) mentioned that lodgepole pine trees with small cone angles were occasional or rare in the southern Rocky Mountains, based on limited data. With the exceptions of populations 1 and 2 in northern Colorado, the results of this study agree with Critchfield's statement. Nevertheless, erect or semi-erect cones (with angles of about 45° to 90°) appeared occasionally in this investigation, and not always in the more northern populations.

The morphological similarities, the absence of strong barriers to gene flow, and the overlapping geographic ranges of jack pine and lodgepole pine suggest a probable, recent divergence of these two taxa. Yeatman (1969) hypothesized that the two trees had a common ancestor distributed in high latitudes prior to Pleistocene glaciation. Glaciation obliterated all jack pine in its current geographic range, and refugia in the Appalachian highlands were probably responsible for the re-expansion of the species (Yeatman, 1967). Lodgepole pine probably survived in refugia in the northwestern part of North America, expanding its range from there after the recession of the glaciers (Critchfield, 1980). The disjunction of the two taxa probably allowed morphological and physiological distinctiveness to develop. Apparently that development has not progressed very far. The certainty which exists in taxonomic classification of the two species indicates that some commonly accepted criteria of what a species is are weak when applied to lodgepole and jack pines.

Some Interpretations of Observed Phenotypic Variation

For most morphological traits, phenotypic expression is only partially controlled by the genotype of an organism. The degree of genetic control of trait expression can vary considerably from trait-to-trait. For instance, tree height and diameter are evidently under strong environmental control, whereas cone orientation angle and serotiny are highly heritable (Wright, 1976; Spurr and Barnes, 1980). Natural selection works through environmental factors to produce changes in the genetic make-up of a population over geographic space

and evolutionary time. As gene frequencies and gene combinations vary population-to-population, the expression of this genetic variability is manifested through phenotypic variation. A given environmental factor or combination of factors may favor one combination of genes in one location, and another combination in another location. Thus, one would not necessarily expect all traits to show similar patterns of variation or to vary together in a parallel manner.

This investigation was patterned along two, large-scale gradients: elevation and latitude. These two gradients encompass numerous factors, including moisture, temperature, topography, length of growing season, and so on. Many of these factors were not measured in this study. Interactions among environmental factors responsible for selection pressures probably differ in kind and intensity in an infinite array throughout the geographic range of lodgepole pine. The diversity of the results illustrated in this study is not surprising considering the relatively few natural populations and few environmental factors included in the analyses.

Several possible explanations exist for large population-to-population variation and often irregular trends in the results. Some of these explanations have been discussed earlier. As alluded to above, the variation observed may have been due to environmental factors and interactions not accounted for. Additionally, patterns of variation at the present time may be due to different selection pressures at an earlier time. It may be possible that some traits, although of adaptive value once in the past, may only be "neutral" now, remaining only because they, or the traits they are genetically linked to, are not selected against. The evolutionary potential that

a population contains in its gene system is released over time as migration occurs, mutations arise, environments change, or new gene combinations are developed (Clausen, 1959).

Small-scale discontinuous patterns of variation may be due to genetic drift (Wright, 1976; Standfield, 1977). In an environmentally heterogeneous region such as the Rocky Mountains, the establishment of genetically different populations separated by relatively short distances seems a reasonable strategy for species adaptation (Mitton et al., 1977). The random nature of environments over short distances could allow genetically and phenotypically diverse populations to develop with no apparent pattern simply by chance dispersal events into isolated areas.

Genetic drift could also explain why, in some cases, within-population variation was less in one sampled stand than in another. If a population or stand happened to be established through random drift, was isolated from other populations, and was small, then one would expect that population to be more homogeneous than a large, widespread population (Mayr, 1977).

Phenotypic plasticity might also help explain some of the variation found among sampled populations. Phenotypic plasticity appears when the same or similar genotypes produce different phenotypes in response to different environments. Such a strategy might be successful in an environmentally heterogeneous region (Rehfeldt, 1979). However, plasticity is not considered to be very important as an adaptive strategy for most forest tree species (Stern and Roche, 1974). Provenance tests or isoenzyme study could reveal more clearly the nature of genetic variation between and within natural populations

and allow a stronger statement on whether or not plasticity could explain some of the morphological variation reported here.

Continuous Versus Discontinuous Variation

One objective of this study was to evaluate the phenotypic variation in several morphological features of lodgepole pine from samples taken throughout the species' range in the central Rocky Mountains. The intent was to use the data collected to either support or refute a case for the existence of clinal variation in Pinus contorta var. latifolia. Several factors influenced the achievement of this objective:

1. The scale of a study such as this one relates directly to the interpretation of the results. For instance, considering the extensive north-to-south range of populations included, a clinal pattern of variation developed by combining population mean values at latitudinal locations could have masked elevational trends. Also, local discontinuities due to soil type, soil moisture, or other environmental factors could have been overlooked.
2. The intensity of sampling over a large geographic area affects conclusions concerning variation patterns (Spurr and Barnes, 1979). If a species is distributed continuously over a large area, and sampling is done only at a few locations (for instance, sampling only two extreme ends of the species' range), then a "discontinuous" interpretation of geographic variation might result; given more intense sampling, the existence of a cline may be a more appropriate conclusion. Gene flow may be restricted between remote parts of the cline

such that local differentiation of natural populations could result. Low intensity sampling which inadvertently includes only the extremes of a cline may lead to erroneous conclusions of discontinuous range-wide variation.

3. Lodgepole pine is a species distributed over a large geographic range, with many breaks in the distribution. Although widely distributed latitudinally, lodgepole pine stands are restricted to moderately high elevation in the Rocky Mountains. These "patches" or "islands" of lodgepole pine may be derived from ancestral populations which were physically connected into a more continuous distribution in the past, before the major uplift of the Rocky Mountains and during a climatically cooler period.

If the time elapsed since the break-up of the continuous pattern has been short, evidence of past gene flow between now disjunct populations may still be apparent. In such a situation, sampling of physically and biologically discontinuous populations could give results interpretable as statistically continuous variation. The question is, then, does statistical continuity imply biological (or genetic) continuity as well? Further testing of genetic variation among populations would help in answering this question. In any case, one should carefully consider the assumptions of the statistical procedures when interpreting the results.

4. Wright (1976) suggested that, instead of relying on connotations of words such as cline, race, or ecotype, it is better to describe the presence or absence of discontinuities and the amount of environmental adaptation when describing a species' variation pattern. Daubenmire (1978) argued that, "... no plants have 'continuous'

distribution in the absolute sense, for suitable habitats do not occur in unbroken series. However, so long as dissemination or pollination allows gene exchange across unfavorable habitats, directly or indirectly, a range may be said to be continuous despite numerous gaps in which the species cannot be found."

Chapter V. Summary and Conclusions

Lodgepole pine is one of the most widely distributed forest tree species in western North America. Although several taxonomic schemes have been developed for the species, it is commonly classified as a single species with three varieties, at the present time: Pinus contorta Doug. ex Loud. var. contorta (shore pine of the Pacific coast), Pinus contorta var. latifolia Engelm. (lodgepole pine of the Rocky Mountains and intermountain region), and Pinus contorta var. murrayana (Grev. and Balf.) Engelm. (Sierra lodgepole pine of the Oregon and Washington Cascade ranges, and the Sierras of norther and central California).

Geographic variation in several phenotypic traits of lodgepole pine was evaluated in this study. Although lodgepole pine has an extensive latitudinal range in the Rocky Mountains, the species' distribution is restricted to a relatively narrow elevational belt immediately below, but usually grading into, forest communities of the subalpine zone, above, and communities of the montane zone, below. Thirty-four natural populations, including a total of 153 individual sample trees, were included in this study. These samples were from throughout the latitudinal and elevational range of lodgepole pine from Montana to southern Colorado.

Measurements were made of phenotypic traits on each tree in the field. Also, cone and needle specimens were collected and

measured in the laboratory. Seed was extracted from cones and seedlings were grown for provenance studies, some of which have been established.

Data from the morphological measurements was analyzed to detect significant differences between natural populations and latitudinal locations (made up of the natural populations at a given latitude), and to reveal geographic trends in phenotypic variation. Results are summarized below:

1. branch angle - Branch angle generally increased from north to south. Southern Colorado populations formed a group with larger branch angles than average. Branch angle was significantly and positively correlated with latitude, longitude, and elevation. Population-to-population variation was statistically continuous, as determined by Duncan's multiple range test. The latter finding was typical for all traits.

2. cone orientation angle - Cone orientation angle was significantly and positively correlated with elevation. Latitudinal trends in cone orientation angle were not strong, but cone orientation angle generally decreased from south to north, and was more variable in the northern populations. Possible introgressive influence from jack pine for this trait was discussed.

3. cone orientation - Similar to cone orientation angle, this trait was significantly correlated with elevation, as well as with latitude. Adaxial cones (pointing in towards the main stem of the tree) were more common at higher elevations and from more southern collections.

4. cone serotiny - The percentage of serotinous cones on a tree varied greatly within and between populations sampled. Two areas of low serotiny appeared in southwest Montana and central Colorado.

5. cone abundance - Cone abundance was significantly correlated with longitude. Most prolific cone production was found in the more eastern collections. Lowest abundance values were from southwest Montana and Wyoming.

6. needle persistence - The number of years that needles persisted on branches was significantly related to latitude, longitude, and elevation of the sample locations. Southern populations had greater persistence values than northern populations.

7. cone length - Cone length was significantly correlated with latitude, longitude, and elevation. Shorter cones were found in collections from northern and western locations and at higher elevations. Southern Colorado populations formed a more-or-less distinct geographic group with larger cone lengths than average.

8. cone width - No significant correlations were found between cone width and any of the geographic variables. A noticeable trend for increasing cone width south-to-north among southern Colorado populations was apparent from the data.

9. cone shape - Cone shape, the ratio of cone length to width, decreased south-to-north, east-to-west, and from high elevations to low elevations.

10. cone volume - Cone volume was significantly lower in samples from higher latitudes, westerly sample locations, and low elevations. Four distinct groupings of populations were illustrated

for this trait. Two groups in Colorado bisected the state, and, in each group, there was an increase in cone volume with increasing latitude. Two other groups of populations in central and northern Montana, and southwest Montana and Wyoming showed a reverse trend.

11. apophysis shape - Whether the basal scales of cone from sampled trees were flat or raised and attenuate was not significantly correlated with any of the geographic variables included in this study. Statistical grouping of population mean values resulted in broad overlap of two, large groups of populations. A slight trend towards flatter apophyses from the northern samples existed.

12. cone weight - Although cone weight was not significantly correlated with any of the geographic variables included in this study, trends in population-to-population variation were similar to those found for cone volume. Two groups of populations in Colorado had increasing cone weight, south-to-north, and two other groups in Wyoming and Montana had an opposite trend.

13. cone specific gravity - Significant correlations were found between cone specific gravity and latitude, longitude, and elevation. Density of cones from sampled trees increased from south-to-north, east-to-west, and from high elevations to low elevations. Population-to-population variation was statistically continuous as with most traits.

14. needle biomass - Needle biomass was significantly related to both latitude and longitude, with needle fascicles from northern and western parts of the sampled range having higher dry weights than fascicles from trees in the south and east.

15. greenhouse seedling height - Under conditions of accelerated growth, seedling progeny from sampled trees demonstrated erratic patterns of growth which could not be related to any of the geographic parameters included in this study. Variation within natural populations was very high so as to mask any geographic patterns. Data from field outplanted provenance trials should provide information to better assess the growth, phenology, and survival potential of the seed sources relative to each other and to a variety of outplanting locations.

Groups of mean values by natural population overlapped broadly for most traits, and continuous variation among populations was illustrated. Further studies on genetic discontinuities of populations would help evaluate whether or not clinal variation patterns were biologically based.

Results from this study generally agree with observations reported in the literature that lodgepole pine is a variable species. Tree-to-tree variation within stands was generally high for most traits, and this may mean that individual tree selection programs may be beneficial for lodgepole pine tree improvement work in the central Rocky Mountains.

In an evolutionary context, lodgepole pine seems to be a relatively young species which may have had a common ancestor with jack pine prior to glaciation in the Pleistocene and uplift of the Rocky Mountains. The large, variable gene pool alluded to above may be responsible for the broad ecologic amplitude of lodgepole pine. Although racial differentiation has been suggested for the species as a

whole, variation within the central Rocky Mountains does not seem to warrant further taxonomic distinctions in this area.

Several avenues of further research should be pursued to obtain an overall understanding of genetic and geographic variation in lodgepole pine in this region. Isoenzyme studies could reveal gene and genotype frequency differences between and within natural populations. Seedling progeny from the present study could provide the raw material for isoenzyme investigations of genetic variation among populations.

Provenance tests, some of which have been established, will provide information relative to growth, survival, and phenology of the several seed source origins.

In general, this investigation of geographic variation in lodgepole pine phenotypes forms a part of a larger effort to gain a better understanding of the species in this region. That increase in knowledge is useful in an economic context when we attempt to manage lodgepole pine for human needs or wishes. The information is also useful in simply increasing the data base from which we attempt to explain how nature works. The human mind always wants to know a little bit more.

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Appendix

Table 45. Summary of sample locations.

Latitudinal Location Number ¹	Natural Population Number ²	Individual Tree Seed Source Number ³	Latitude (°N)	Longitude (°W)	Elevation (meters)
1	1	co-1-1	40.6	105.4	2378
		-2			
		-3			
		-4			
		-5			
	2	co-2-1	40.6	105.5	2588
		-2			
		-3			
		-4			
		-5			
	16	co-16-1	40.6	105.5	2805
		-2			
		-3			
		-4			
		-5			
2	3	co-3-1	37.6	105.2	2805
		-2			
		-3			
		-4			
		-5			
3	4	co-4-1	38.4	105.4	2744
		-2			
		-3			
		-4			
		-5			
	5	co-5-1	38.5	105.4	3049
		-2			
		-3			
		-4			
		-5			
	6	co-6-1	38.5	106.3	3338
		-2			
		-3			
		-4			
		-5			

Table 45. (continued)

Latitudinal Location Number	Natural Population Number	Individual Tree Seed Source Number	Latitude (°N)	Longitude (°W)	Elevation (meters)
4	7	co-7-1	38.8	106.3	3003
		-2			
		-3			
		-4			
		-5			
	8	co-8-1	38.8	106.4	3079
		-2			
		-3			
		-4			
	9	co-9-1	38.8	106.4	3506
		-2			
		-3			
5	10	co-10-1	39.4	106.3	3178
		-2			
		-3			
		-4			
		-5			
	11	co-11-1	39.4	106.3	3201
		-2			
		-3			
		-4			
		-5			
	12	co-12-1	39.5	106.4	2652
		-2			
		-3			
		-4			
		-5			
6	13	co-13-1	40.3	106.1	2652
		-2			
		-3			
		-4			
		-5			

Table 45. (continued)

Latitudinal Location Number	Natural Population Number	Individual Tree Seed Source Number	Latitude (° N)	Longitude (° W)	Elevation (meters)
6	14	co-14-1	40.4	106.1	2835
		-2			
		-3			
		-4			
		-5			
	15	co-15-1	40.4	106.1	2683
		-2			
		-3			
		-4			
		-5			
7	17	mt-1-1	46.8	110.8	1875
		-2			
		-3			
		-4			
	18	mt-2-1	46.8	110.7	2165
		-2			
		-3			
		-4			
	19	mt-3-1	47.0	110.8	1561
		-2			
		-3			
		-4			
8	20	mt-4-1	48.3	113.4	1555
		-2			
		-3			
		-4			
		-5			
	21	mt-5-1	48.3	113.4	1524
		-2			
		-3			
		-4			
		-5			

Table 45. (continued)

Latitudinal Location Number	Natural Population Number	Individual Tree Seed source Number	Latitude (°N)	Longitude (°W)	Elevation (meters)
9	22	mt-6-1	46.7	114.5	1372
		-2			
		-3			
		-4			
10	23	mt-7-1	45.7	114.0	2134
		-2			
		-3			
		-4			
	24	mt-8-1	45.7	114.0	2210
		-2			
		-3			
		-4			
	25	mt-9-1	45.7	113.9	1951
		-2			
		-3			
		-4			
11	26	mt-10-1	45.4	111.2	1783
		-2			
		-3			
		-4			
	27	mt-11-1	44.8	111.2	2012
		-2			
		-3			
		-4			
	28	mt-12-1	44.9	111.4	2287
		-2			
		-3			
		-4			
12	29	wy-1-1	44.8	107.4	2287
		-2			
		-3			
		-4			

Table 45. (continued)

Latitudinal Location Number	Natural Population Number	Individual Tree Seed Source Number	Latitude (°N)	Longitude (°W)	Elevation (meters)
12	30	wy-2-1	44.8	107.5	2591
		-2			
		-3			
		-4			
		-5			
13	31	wy-3-1	42.7	108.9	2439
		-2			
		-3			
		-4			
		-5			
	32	wy-4-1	41.7	108.9	2591
		-2			
		-3			
		-4			
		-5			
	33	wy-5-1	42.7	108.9	2920
		-2			
	34	wy-5a-1	42.6	108.8	2668
		-2			
		-3			

¹ See Figure 2.

² See Figure 3.

³ The first two letters correspond to state: co = Colorado, wy = Wyoming, mt = Montana. The first number after the state designation corresponds to the stand visited in consecutive order within a state (1 = first, 2 = second, and so on). The last set of digits corresponds to the tree number within a stand. Example: co-5-3 represents the third tree in the fifth stand visited in Colorado.

Table 46. Sample field data form - obverse.

CSFS - Tree Improvement Project
Field Data Sheet

Study Title: Lodgepole Pine Variation Observers: _____

Collection No. _____ Seed Zone No. _____ Date _____

Site Data

Location: T _____ R _____ Sec. _____ ° _____ PM 1/4 Sec. _____ "40" _____

Latitude _____ Longitude _____ Elevation _____

Locality: _____ miles _____ from _____
direction _____

Topography: Bottomland Slope Upland Ridgetop (Circle)

Aspect: North East South West North (Circle)

1 2 3 4 5 6 7 8

Slope: _____ percent Forest type: _____

(all tree species present)

Top soil texture: Gravel Sand Loam Clay Origin: _____

1 2 3 4 5 6 7 (Circle)

Tree Data

Species: _____

Tree Number: _____

Height: _____

D.B.H.: _____

Age: _____

Other phenotypic features: _____

Materials collected:

Leaves _____ Flowers _____

Twigs _____ Fruit _____

Bark _____ Seeds _____

Mark plot location in quarter-section square on _____

Remarks _____

Table 47. Sample field data form - reverse.

ADDITIONAL FIELD INFORMATION - VARIATION STUDY

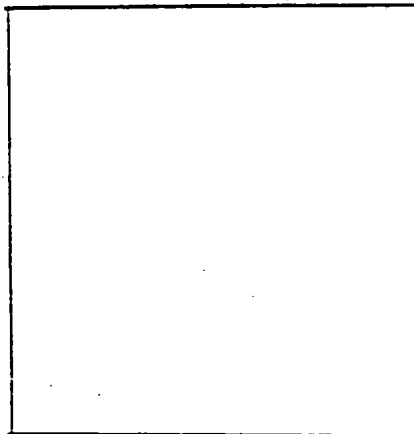
Tree Number _____

Phenotypic Measures

1. Branch Angle:
(degrees) _____ / _____
mean
2. Cone Orientation Angle:
(degrees) _____ / _____
mean
3. Needle Persistence
(years) _____ / _____
mean
4. Type of Cone Orientation: (check one)

abaxialadaxial

5. Cone Serotiny:
(circle) 1 2 3 4 5
0-10% 10-25% 25-50% 50-75% 75-100% serotinous
6. Cone Abundance:
(circle) 1 2 3 4 5
few many

Stand Climate Data (complete in office)

Stand or Tree Photo

Negative Number: _____

Table 48. Example data form for laboratory cone measurements.

Lodgepole Pine Variation
Cone and Seed Data Form

Population _____ Notes by _____
Tree Number _____ Date _____
Collection _____

Morphology

1/				Apophyses		2/
Cone No.	Length	Width	Shape	Length	Width	Knobiness
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

1/ Ratio of Length to Width

2/ Enobiness classification:

1	2	3	4	5
Flat				Attenuate

Cone Weight

Date: _____ Weight of sample _____ (=5.) (10 cones)
 ÷ _____ sample size
 = _____ Mean Cone Weight

Cone Volume

Cone Volume Volume of H_2O + cone sample (10 cones): _____ ml.
 Date: _____ Volume of H_2O : - _____ ml.
 Cone volume : = _____
 Divided by sample size _____ = _____ Mean Vol.

Table 49. Example data form for greenhouse seedling measurements.

Seed Germination and Seedling Observation Data

Project 2260

Study Title Lodgement Line:

Source No.	Block No.	Plot No.	Treatment
------------	-----------	----------	-----------

<u>Date</u>	<u>Down</u>	<u>Outplanted</u>
12-1-68	10	10
12-2-68	10	10
12-3-68	10	10
12-4-68	10	10
12-5-68	10	10
12-6-68	10	10
12-7-68	10	10
12-8-68	10	10
12-9-68	10	10
12-10-68	10	10
12-11-68	10	10
12-12-68	10	10
12-13-68	10	10
12-14-68	10	10
12-15-68	10	10
12-16-68	10	10
12-17-68	10	10
12-18-68	10	10
12-19-68	10	10
12-20-68	10	10
12-21-68	10	10
12-22-68	10	10
12-23-68	10	10
12-24-68	10	10
12-25-68	10	10
12-26-68	10	10
12-27-68	10	10
12-28-68	10	10
12-29-68	10	10
12-30-68	10	10
12-31-68	10	10

[illegible]

2 Data Type:

- 0 = Germination, normal
- 1 = Germination, abnormal
- 2 = Ht. (mm)
- 3 = Dia. (mm, 5 mm from soil level)
- 4 = Other (specify) _____
- 5 = Other (specify) _____

2/ Condition Codes:

- 0 = Dead
1 = Germinating
2 = Active Growth
3 = Bud Set

(cotyledon counts, needle counts, etc.)

5 - Other (specify) _____

Table 50. Format for one-way analysis of variance for testing differences between 34 natural populations or 13 latitudinal locations.

Source of Variation	Degrees of Freedom	Estimated Mean Square
Between Groups	$k-1$	$\sigma_w^2 + n\sigma_b^2$
Within Groups (residual)	$n-k$	σ_w^2
Total	$n-1$	

Where:

k = Number of groups in analysis; either 34 natural populations or 13 latitudinal locations.

n = Total number of observations.

σ_w^2 = Residual variance.

σ_b^2 = Variance among groups.

Table 51. Summary of analysis of variance results for the 34 natural populations based on morphological traits.

Morphological Trait	F (df: 33, 119)	Significance
branch angle	3.940	< 0.001
cone orientation angle	3.564	< 0.001
needle persistence	9.948	< 0.001
cone orientation	2.594	< 0.001
cone serotiny	3.512	< 0.001
cone abundance	3.555	< 0.001
cone length	3.160	< 0.001
cone width	2.132	0.002
cone shape	1.573	0.041
cone volume	4.173	< 0.001
apophysis shape	1.341	0.129
cone weight	3.868	< 0.001
cone specific gravity	3.370	< 0.001
needle biomass	6.710	< 0.001
greenhouse seedling height	1.216	0.222

Table 52. Summary of analysis of variance results for the 13 latitudinal locations based on morphological traits.

Morphological Trait	F(df: 12, 140)	Significance
branch angle	4.806	< 0.001
cone orientation angle	6.200	< 0.001
needle persistence	10.913	< 0.001
cone orientation	5.244	< 0.001
cone serotiny	6.063	< 0.001
cone abundance	4.813	< 0.001
cone length	6.434	< 0.001
cone width	3.800	< 0.001
cone shape	2.747	0.002
cone volume	7.616	< 0.001
apophysis shape	2.221	0.014
cone weight	6.201	< 0.001
cone specific gravity	6.616	< 0.001
needle biomass	4.557	< 0.001
greenhouse seedling height	1.239	0.263

Figure 34a. Representative cones from Colorado collection sites
1, 2, 3, 4, and 5.

Figure 34b. Representative cones from Colorado collection sites
6, 7, 8, 9, and 10.

COLORADO

CO-

COLLECTION SITE N°

TREE N°	COLLECTION SITE N°				
	1	2	3	4	5
1					
2					
3					
4					
5					

COLORADO

CO-

COLLECTION SITE N°

TREE N°	COLLECTION SITE N°				
	6	7	8	9	10
1					
2					
3					
4					
5					

Figure 34c. Representative cones from Colorado collection sites 11, 12, 13, 14, 15, and 16.

Figure 35 Representative cones from Wyoming collection sites 1, 2, 3, 4, 5, and 5a.

COLORADO CO-

COLLECTION SITE NO

TREE NO	11	12	13	14	15	16
1						
2						
3						
4						
5						

WYOMING WY-

COLLECTION SITE NO

TREE NO	1	2	3	4	5	5A
1						
2						
3						
4						
5						

Figure 36a. Representative cones from Montana collections sites
1, 2, 3, 4, 5, and 6.

Figure 36b. Representative cones from Montana collection sites
7, 8, 9, 10, 11, and 12.

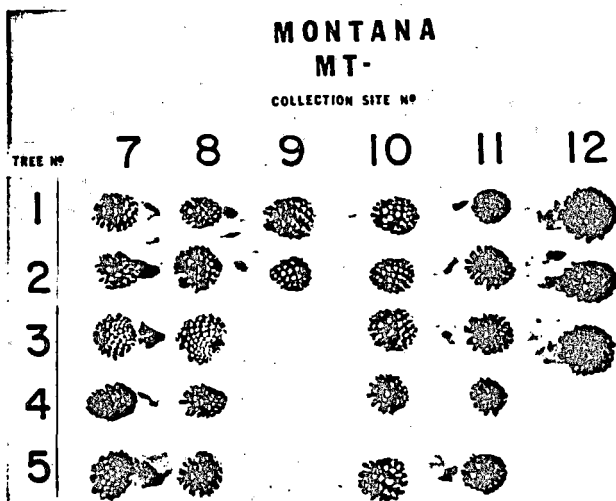
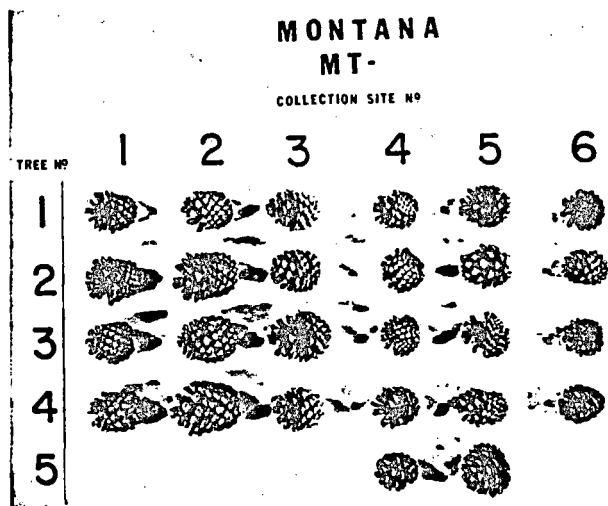


Figure 37. Layout of the lodgepole pine provenance test plantation at the Colorado State Forest near Gould, Colorado.

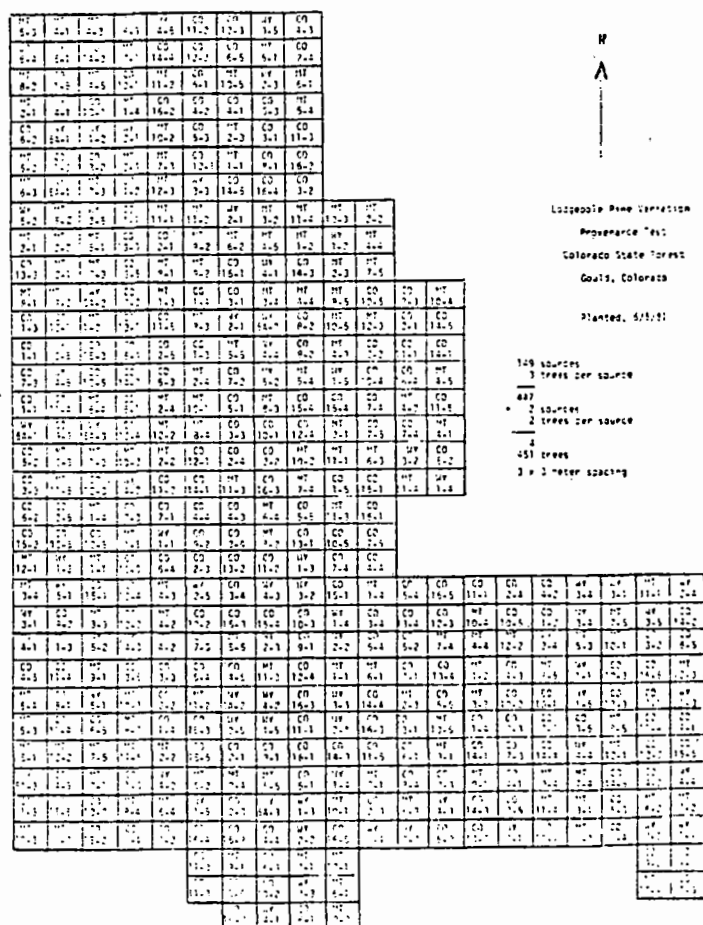


Figure 38a. Layout of block I of the lodgepole pine provenance test
plantation at the Fraser Experimental Forest (USDA
Forest Service) near Fraser, Colorado.

BLOCK I

CO-2-1	CO-13-1	MT-11-4	CO-2-3	WY-4-4	MT-10-2	WY-4-2	CO-8-2	CO-1-3	MT-7-1	WY-1-1	CO-15-5	CO-7-1	CO-8-1	CO-10-1	WY-5A-1
CO-12-3	CO-15-2	CO-8-3	WY-3-4	CO-2-4	CO-3-1	MT-11-1	CO-1-5	MT-3-2	CO-11-3	MT-7-5	CO-5-5	MT-2-2	MT-8-2	CO-15-1	CO-13-5
MT-5-2	CO-3-2	CO-13-5	CO-8-1	MT-11-3	CO-2-5	MT-8-4	MT-4-2	CO-7-1	CO-6-3	WY-1-3	MT-8-5	WY-1-5	MT-3-3	WY-1-1	WY-4-3
CO-10-3	CO-7-3	CO-7-4	CO-5-1	WY-3-4	CO-7-5	CO-14-3	WY-5-1	MT-5-2	CO-16-4	MT-1-4	CO-1-4	MT-5-4	MT-10-3	CO-8-4	WY-3-1
CO-14-2	CO-2-2	CO-11-2	CO-16-4	CO-10-1	MT-3-1	MT-8-3	CO-10-2	MT-11-1	CO-7-2	MT-1-2	MT-12-3	WY-2-1	MT-5-1	WY-5-1	CO-5-2
MT-12-3	WY-5-1	CO-13-1	WY-1-2	MT-7-3	CO-1-3	CO-2-5	CO-14-5	MT-8-1	CO-12-3	MT-5-5	MT-1-1	CO-16-2	CO-3-2	MT-11-1	CO-15-2
CO-16-2	MT-10-1	CO-4-2	MT-7-4	CO-16-5	CO-4-5	CO-1-5	WY-2-3	WY-2-1	WY-4-1	MT-3-4	CO-4-3	CO-12-5	CO-4-1	CO-3-4	MT-5-4
WY-1-1	MT-1-3	WY-2-2	WY-2-2	CO-15-5	CO-15-4	CO-15-5	CO-12-2	CO-11-5	CO-1-1	CO-7-4	CO-7-5	MT-5-1	MT-8-5	CO-1-4	MT-4-1
MT-7-3	MT-4-1	MT-4-5	MT-12-1	CO-7-2	CO-12-4	CO-4-2	CO-4-4	CO-3-1	CO-15-3	CO-2-1	MT-12-1	CO-10-5	CO-13-3	WY-4-2	MT-3-2
CO-13-5	WY-4-5	CO-5-3	MT-9-1	MT-11-2	MT-11-3	CO-11-5	MT-11-5	CO-8-3	WY-3-2	MT-2-1	CO-1-3	MT-4-3	CO-6-2	CO-3-5	WY-3-3
MT-8-4	WY-1-4	CO-7-2	MT-6-1	CO-3-5	MT-1-4	CO-2-1	WY-3-5	CO-7-3	CO-10-3	WY-2-4	CO-1-1	MT-7-1	MT-6-2	CO-13-1	MT-4-3
CO-12-2	CO-13-3	MT-4-4	CO-14-4	WY-2-5	MT-8-5	MT-1-3	WY-4-3	CO-1-1	MT-9-1	MT-8-3	CO-16-1	CO-5-1	MT-11-5	MT-2-1	MT-1-1
CO-5-4	CO-11-3	CO-11-4	MT-5-2	CO-14-3	CO-8-1	CO-1-2	CO-15-3	MT-12-1	MT-10-2	MT-4-2	MT-5-4	WY-1-4	MT-10-1	CO-11-1	MT-3-1
MT-3-1	CO-16-5	CO-15-4	WY-2-5	CO-2-3	MT-11-2	CO-13-2	WY-5-2	WY-1-2	CO-13-2	CO-14-1	CO-16-3	CO-5-5	CO-4-5	CO-6-2	WY-5A-2
CO-4-5	CO-4-3	CO-13-4	MT-2-4	CO-13-4	CO-4-4	CO-7-1	WY-5A-2	WY-3-2	MT-5-5	CO-5-4	WY-5A-3	MT-6-1	WY-5A-1	CO-6-2	WY-4-5
MT-4-5	CO-4-3	MT-2-2	MT-6-2	MT-6-3	MT-1-2	WY-5A-1	WY-1-3	CO-7-4	MT-5-5	CO-3-3	CO-6-4	CO-14-5	CO-8-3	MT-2-3	MT-11-3
CO-5-5	WY-3-3	CO-10-4	CO-3-2	CO-2-4	CO-14-4	MT-3-4	CO-10-1	MT-8-1	MT-5-3	CO-10-4	CO-16-1	CO-2-5	MT-2-4	WY-2-3	MT-12-3
CO-11-2	MT-6-4	MT-11-4	CO-12-1	WY-4-3	MT-1-4	MT-11-2	MT-7-5	MT-9-2	MT-3-3	CO-11-2	CO-16-4	CO-6-5	WY-4-1	WY-2-2	CO-12-1
CO-3-4	CO-15-4	WY-2-4	CO-1-2	MT-10-2	CO-12-5	CO-6-4	MT-7-5	CO-15-1	MT-9-2	CO-11-3	CO-14-4	MT-10-3	CO-2-3	MT-2-2	WY-3-2
CO-3-3	WY-5-2	MT-7-1	MT-4-5	CO-5-2	MT-6-3	CO-12-5	CO-10-5	CO-4-3	MT-5-1	MT-8-1	WY-1-5	WY-2-3	MT-1-2	CO-5-2	MT-8-2
CO-14-1	WY-5A-3	CO-8-1	CO-15-3	CO-10-4	MT-4-4	CO-14-3	CO-2-2	MT-12-2	CO-14-2	MT-2-4	CO-11-5	MT-10-1	MT-10-5	WY-5-2	CO-5-3
MT-8-2	MT-5-3	WY-1-3	WY-1-4	MT-6-1	WY-5A-3	CO-4-4	CO-16-2	MT-11-4	CO-14-2	CO-7-5	WY-4-4	MT-10-5	CO-9-2	MT-3-3	WY-2-5
MT-11-5	CO-1-4	CO-8-2	CO-5-3	CO-6-1	CO-10-2	CO-12-1	CO-3-5	MT-4-2	CO-5-1	CO-15-1	CO-9-2	WY-4-5	WY-2-4	CO-1-2	MT-12-3
MT-6-4	MT-5-3	MT-10-3	MT-7-3	WY-3-4	CO-11-4	CO-5-4	MT-12-2	WY-5A-2	CO-2-2	MT-2-3	CO-3-4	CO-12-4	MT-6-4	MT-4-4	CO-4-1
CO-6-5	CO-6-1	MT-4-1	CO-2-4	MT-10-5	MT-6-2	CO-9-2	CO-16-5	WY-3-5	MT-7-4	WY-3-1	MT-2-3	MT-3-4	CO-7-3	MT-6-3	CO-6-3
CO-16-3	CO-11-1	CO-6-5	CO-13-4	MT-2-1	CO-1-5	WY-4-4	MT-1-1	CO-11-1	CO-16-4	CO-6-1		CO-12-2	CO-3-3	WY-1-5	MT-12-2
CO-4-1	CO-10-3	CO-12-4	WY-2-1	WY-4-2	CO-8-4	CO-14-1	WY-3-1	CO-12-3	MT-4-3	MT-8-3	WY-9-2	MT-3-2	WY-4-1	CO-11-4	CO-10-2
CO-6-4	MT-9-1	MT-8-4	WY-1-2	CO-3-4	CO-13-3	CO-16-1	CO-6-3	MT-1-3	MT-7-4	CO-3-1	CO-14-5	CO-15-2	CO-13-2	CO-10-5	WY-3-3

Figure 38b. Layout of block II of the lodgepole pine provenance test plantation at the Fraser Experimental Forest (USDA Forest Service) near Fraser, Colorado.

BLOCK II

MT-3-3	CO-2-5	CO-11-4	CO-10-3	CO-4-5	CO-6-4	CO-14-4	MT-5-5	CO-14-2	CO-13-3	CO-13-4	CO-14-1	MT-3-3	MT-9-2	CO-2-4	CO-13-1
MT-3-1	MT-8-4	MT-2-4	MT-3-1	CO-11-3	CO-12-3	CO-15-1	CO-5-1	MT-2-1	WY-1-2	CO-16-2	CO-3-4	MT-8-4	MT-11-5	WY-1-1	CO-14-4
CO-11-1	MT-1-2	MT-5-4	CO-8-4	CO-16-1	MT-2-3	CO-6-4	CO-10-4	MT-3-4	CO-14-3	CO-10-1	CO-4-3	MT-7-1	MT-7-4	MT-1-4	WY-3-3
CO-15-2	CO-13-4	CO-16-3	MT-5-1	MT-2-2	CO-15-2	MT-4-5	CO-6-5	MT-10-1	CO-10-1	WY-4-1	CO-6-3	CO-7-3	CO-10-5	CO-2-5	MT-2-4
WY-3-1	MT-11-4	MT-1-1	CO-7-5	CO-6-1	MT-10-2	WY-1-3	CO-10-2	MT-5-2	WY-3-5	CO-7-1	WY-4-1	CO-13-2	MT-4-5	CO-3-5	MT-4-1
CO-4-3	WY-4-3	MT-12-3	CO-3-2	WY-3-5	MT-7-3	WY-4-2	MT-5-3	CO-3-3	MT-6-2	WY-4-4	MT-7-1	CO-1-5	MT-8-3	MT-7-3	CO-3-5
MT-5-2	CO-10-2	MT-7-5	CO-5-5	MT-8-5	MT-4-5	CO-3-1	CO-12-3	MT-9-1	CO-6-5	CO-6-4	MT-5-4	MT-4-4	CO-5-3	MT-10-5	WY-5-1
CO-8-2	MT-4-3	MT-5-3	CO-6-1	CO-4-1	WY-2-5	MT-3-3	CO-4-4	CO-12-2	MT-10-3	WY-5-2	CO-3-3	CO-9-2	CO-3-4	CO-10-2	MT-1-4
WY-5-2	WY-3-2	MT-10-5	MT-9-1	MT-4-4	CO-14-4	CO-11-5	CO-13-2	CO-1-4	CO-2-4	WY-5A-1	WY-2-3	MT-11-3	MT-8-3	WY-4-4	MT-12-2
CO-16-3	CO-12-1	CO-3-1	CO-7-1	CO-3-5	WY-4-3	CO-11-1	MT-10-3	CO-13-2	CO-5-2	MT-5-1	MT-5-2	MT-6-3	MT-6-1	MT-6-2	CO-7-3
WY-5A-1	CO-5-3	MT-12-1	MT-9-2	MT-3-2	MT-6-1	CO-1-4	MT-9-2	CO-7-3	CO-12-3	CO-15-4	MT-8-5	CO-4-2	CO-4-5	WY-3-5	CO-12-4
WY-5A-3	WY-2-2	CO-7-1	CO-2-5	CO-12-5	CO-10-3	CO-5-5	WY-1-1	CO-8-3	WY-3-2	CO-16-5	WY-3-4	WY-4-1	CO-1-4	WY-5A-2	CO-16-4
MT-11-3	MT-10-1	CO-8-3	CO-1-5	WY-3-3	CO-1-2	MT-4-1	CO-3-2	CO-7-2	CO-11-3	WY-1-2	CO-2-4	WY-1-5	MT-10-2	CO-13-4	MT-5-4
MT-7-4	MT-4-4	CO-6-1	MT-3-4	MT-11-5	WY-1-5	WY-4-4	MT-6-4	CO-10-5	MT-3-4	CO-5-4	MT-4-2	CO-5-1	CO-1-1	WY-5A-3	CO-5-1
WY-5A-2	CO-16-5	CO-8-1	CO-13-1	WY-2-3	CO-8-1	CO-4-2	WY-2-5	MT-5-5	WY-4-5	MT-4-2	MT-10-1	CO-14-2	CO-11-3	CO-16-1	CO-2-1
CO-16-5	CO-11-2	WY-3-3	MT-12-3	CO-12-1	CO-9-2	CO-16-2	MT-2-3	CO-6-2	MT-3-2	WY-2-5	MT-12-5	CO-15-3	CO-3-3	CO-14-1	WY-2-2
CO-11-4	CO-15-4	CO-14-3	MT-6-4	CO-7-2	CO-3-2	WY-3-1	CO-2-3	MT-2-4	MT-5-3	WY-5A-2	CO-12-4	CO-15-2	CO-7-4	WY-5A-1	MT-11-1
MT-5-5	MT-7-1	WY-2-1	MT-7-4	CO-15-3	CO-5-4	MT-7-5	MT-12-2	CO-11-1	MT-8-4	CO-7-5	CO-14-5	CO-12-2	CO-2-2	MT-8-2	MT-6-1
WY-4-2	MT-6-3	WY-1-3	CO-6-2	MT-4-3	CO-12-5	MT-11-4	MT-4-2	MT-12-1	CO-15-4	MT-8-1	CO-7-4	MT-1-1	CO-10-4	CO-11-5	CO-3-1
WY-2-1	WY-1-1	CO-5-5	CO-4-4	MT-10-2	MT-1-3	MT-1-1	WY-4-2	MT-9-1	CO-2-3	CO-13-5	CO-13-3	CO-1-1	CO-14-2	WY-3-2	MT-8-2
WY-5-1	CO-6-5	CO-15-3	CO-5-3	WY-1-4	CO-4-1	CO-1-3	CO-13-5	CO-5-2	MT-7-3	CO-12-5	WY-1-4	MT-11-1	MT-2-3	CO-10-3	MT-11-2
CO-1-2	CO-4-3	CO-2-2	CO-10-5	CO-2-2	MT-11-4	CO-10-4	CO-14-5	CO-15-5	CO-7-4	MT-1-2	CO-16-1	CO-16-4	CO-8-4	MT-3-1	CO-11-2
	MT-2-2	MT-8-3	MT-1-3	MT-7-5	CO-15-5	MT-10-3	MT-11-2	MT-8-1	WY-2-4	WY-5-2	CO-9-2	CO-6-3	CO-6-2	CO-1-3	CO-14-5
CO-13-5	CO-2-1	CO-12-1	CO-8-2	CO-8-3	CO-2-3	MT-6-2	CO-16-2	WY-3-1	MT-1-3	MT-6-3	MT-3-2	CO-3-4	CO-4-1	WY-2-2	CO-6-3
WY-1-2	CO-14-3	MT-4-1	MT-11-2	CO-1-2	MT-4-3	WY-2-4	MT-2-1	WY-2-3	CO-4-2	MT-2-2	CO-5-2	WY-1-3	MT-11-5	CO-5-4	CO-8-4
WY-4-5	CO-7-5	CO-4-4	MT-10-5	CO-14-1	WY-2-4	CO-10-1	WY-4-5	CO-4-5	WY-5A-3	WY-2-1	MT-12-1	CO-11-2	MT-1-2	CO-8-1	WY-1-5
CO-1-5	MT-12-2	CO-13-1	MT-8-5	CO-16-4	MT-8-2	CO-8-2	CO-7-2	CO-2-1	CO-1-3	MT-11-3	CO-11-4	WY-3-4	CO-1-1	MT-11-1	MT-8-1
MT-2-1	WY-4-3	CO-15-1	CO-12-2	MT-6-4	CO-13-3	WY-5-1	WY-3-4	WY-1-4	CO-15-1	CO-12-4	MT-1-4	CO-15-5	CO-11-5	MT-5-1	CO-16-3

Figure 38c. Layout of block III of the lodgepole pine provenance test plantation at the Fraser Experimental Forest (USDA Forest Service) near Fraser, Colorado.

BLOCK III

MT-3-2	CO-6-4	CO-4-5	CO-4-1	CO-5-2	CO-6-5	MT-8-4	MT-9-1	MT-3-1	CO-6-2	MT-4-5	CO-3-2	MT-11-5	MT-6-1	WY-1-1	MT-6-2
CO-2-5	MT-1-1	WY-3-4	MT-1-3	CO-6-3	MT-4-2	CO-8-1	CO-5-5	CO-5-1	MT-2-4	MT-2-4	CO-4-5	CO-6-1	CO-4-1	MT-2-3	CO-3-4
CO-4-4	CO-7-1	MT-4-4	CO-2-4	WY-2-5	CO-5-3	CO-14-4	MT-5-5	CO-5-5	CO-6-5	MT-9-2	WY-2-1	MT-2-1	CO-9-2	CO-5-2	WY-1-5
CO-1-2	MT-9-2	CO-5-4	MT-5-3	MT-8-3	CO-3-1	WY-1-2	MT-11-5	CO-3-4	CO-2-4	WY-1-1	MT-5-4	WY-1-1	MT-1-2	CO-11-4	MT-11-3
MT-4-2	WY-3-5	MT-7-3	CO-6-2	CO-4-2	MT-10-5	CO-15-2	MT-7-5	MT-5-2	MT-8-1	CO-4-3	CO-6-3	CO-7-5	CO-12-1	CO-12-5	MT-8-5
CO-13-2	WY-4-5	MT-11-5	MT-6-2	CO-2-3	MT-7-3	CO-15-5	CO-1-4	WY-4-3	MT-12-1	CO-8-3	CO-2-1	WY-5A-3	MT-11-2	CO-13-5	CO-6-3
CO-14-1	CO-13-4	MT-5-1	CO-1-5	MT-11-1	MT-5-2	MT-7-5	CO-11-4	MT-7-3	MT-1-4	CO-13-4	CO-7-3	CO-12-4	WY-1-5	WY-3-3	CO-7-5
CO-11-2	CO-8-4	MT-1-3	CO-15-1	MT-5-5	WY-1-4	CO-11-3	WY-3-4	MT-11-2	CO-8-3	CO-5-4	MT-6-1	MT-6-4	WY-3-1	MT-7-4	WY-4-1
CO-14-5	CO-6-1	WY-1-2	MT-12-2	MT-4-5	CO-2-2	CO-5-3	CO-14-2	WY-1-3	CO-3-2	CO-2-1	MT-1-3	MT-8-2	CO-16-1	MT-10-2	CO-8-2
MT-8-5	MT-12-1	MT-10-5	WY-1-3	WY-3-3	MT-10-1	MT-2-4	CO-15-2	MT-6-1	CO-7-4	WY-4-1	CO-10-3	CO-13-5	MT-4-4	MT-5-3	MT-2-3
MT-6-3	WY-2-3	CO-12-5	WY-3-4	MT-1-2	MT-1-2	CO-12-4	WY-1-3	CO-8-4	CO-11-5	CO-12-3	CO-5-5	CO-15-2	CO-16-3	CO-14-2	CO-12-2
MT-12-2	CO-1-1	CO-14-4	CO-12-4	WY-5A-1	WY-5A-1	CO-3-1	WY-2-4	MT-2-1	WY-3-2	CO-16-2	WY-2-1	CO-16-4	MT-4-5	CO-3-1	WY-5-2
CO-10-4	MT-7-4	CO-7-3	CO-6-4	CO-4-1	CO-11-2	MT-4-3	MT-5-4	MT-9-1	WY-3-1	WY-5A-3	CO-15-4	WY-4-3	WY-5A-2	WY-5A-3	MT-11-1
CO-14-3	CO-14-3	MT-3-1	WY-5-2	CO-1-1	CO-7-2	MT-2-2	MT-10-3	CO-6-4	CO-1-2	CO-3-3	WY-5A-2	MT-5-5	CO-13-1	MT-1-4	CO-13-3
CO-3-3	WY-1-4	CO-14-2	MT-8-4	MT-7-5	CO-12-3	CO-16-5	WY-4-4	MT-4-1	CO-16-4	MT-1-1	CO-15-4	WY-2-3	CO-5-1	CO-7-4	CO-4-4
MT-10-3	CO-2-2	CO-10-2	WY-2-2	CO-12-5	CO-13-2	WY-5-1	CO-10-1	WY-3-5	MT-6-4	MT-11-4	WY-4-1	MT-4-3	CO-1-3	WY-2-2	MT-8-1
CO-13-3	CO-15-1	MT-3-4	WY-4-2	CO-15-3	WY-3-5	CO-4-2	MT-3-3	WY-2-4	CO-5-1	MT-6-3	WY-5-1	CO-10-5	CO-15-3	MT-8-4	CO-15-4
CO-7-1	MT-5-4	WY-2-5	CO-14-5	MT-2-2	WY-2-2	MT-6-2	CO-3-5	CO-12-2	WY-5-1	WY-4-3	MT-4-1	MT-3-2	CO-15-5	CO-8-2	MT-3-4
CO-9-2	MT-7-1	CO-13-4	CO-10-1	CO-16-3	MT-7-1	MT-5-1	CO-4-3	CO-16-3	CO-12-2	CO-1-5	CO-11-5	CO-16-5	WY-4-5	WY-4-4	CO-10-2
MT-1-1	CO-15-2	MT-11-1	MT-10-2	CO-13-2	CO-10-4	WY-2-1	CO-16-5	CO-2-3	MT-8-3	MT-5-3	MT-10-2	MT-10-1	CO-7-2	CO-6-5	CO-1-5
MT-6-3	MT-3-1	CO-2-4	MT-8-1	CO-7-2	CO-13-5	MT-8-2	CO-16-4	CO-15-5	WY-5-2		MT-7-4	CO-3-3	CO-2-2	MT-12-3	WY-5A-1
WY-4-4	CO-16-1	WY-4-2	MT-3-3	CO-11-1	CO-1-3	WY-3-2	MT-2-3	CO-16-1	WY-1-2	WY-3-1	CO-2-3	CO-1-4	MT-3-2	CO-1-4	MT-5-2
MT-5-1	CO-3-4	WY-5A-2	MT-12-3	CO-8-1	MT-2-2	MT-10-5	MT-1-4	CO-1-2	CO-14-5	MT-9-2	MT-3-3	CO-2-1	MT-11-3	CO-10-5	CO-2-5
WY-1-5	MT-4-4	CO-3-5	MT-8-3	MT-4-1	CO-14-3	CO-1-3	CO-10-3	CO-16-2	MT-4-2	CO-11-1	MT-11-4	CO-13-1	WY-2-4	MT-11-3	WY-2-3
WY-2-5	CO-4-3	CO-11-4	CO-7-3	WY-1-4	CO-4-4	MT-12-1	CO-12-1	CO-13-2	CO-14-4	CO-10-2	MT-12-3	CO-11-5	MT-8-2	MT-4-3	MT-2-1
WY-3-3	CO-16-2	CO-13-3	CO-14-1	MT-11-2	WY-4-5	CO-3-5	CO-2-5	MT-12-2	MT-10-3	MT-10-1	MT-8-5	WY-3-2	CO-5-2	CO-5-3	CO-15-3
CO-6-1	CO-12-1	CO-8-4	CO-11-3	WY-4-2	MT-11-4	CO-10-5	CO-8-2	CO-11-2	CO-14-1	MT-3-4	CO-12-3	CO-9-2	MT-6-4	CO-4-5	CO-11-1
MT-7-1	CO-11-3	CO-1-1	CO-4-2	CO-10-3	CO-3-1	CO-8-3	CO-7-4	CO-7-5	CO-10-4	CO-7-1	CO-10-1	CO-6-2	MT-9-1	CO-3-2	CO-5-4